



Due to limited time, development of this package is currently suspended.

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1 Basic Introduction

To this day, there are many drawing packages based on `tikz`, each with different purposes and found across various fields. Some packages already provide interfaces similar to `ztikz`, including:

- Common `Ti $\kern-0.05em k$ Z` command wrappers: `tzplot`;
- `Ti $\kern-0.05em k$ Z` package for 3D drawing: `tikz-3dplot`;
- New general-purpose drawing package `luadraw`, which natively supports 3D drawing, currently supporting only `Lua $\TeX$` ;
- PSTricks-based (special) function drawing package: `pst-func`;
- Package for caching compilation results: `robust-externalize`;
- ...

If you find that the `ztikz` package does not meet your needs, you might try the packages mentioned above, or directly use the commands and libraries provided by the original `tikz` package. There are also rich `Ti $\kern-0.05em k$ Z` resources online, such as the `Ti $\kern-0.05em k$ Z` drawing website – `Ti $\kern-0.05em k$ Z Example`, which contains many drawing examples and their corresponding code.

However, the interfaces provided by the above packages are not very uniform and I am not used to them, which is why I decided to develop the `ztikz` package. The command format of `zTi $\kern-0.05em k$ Z` basically follows the naming conventions of functions in Mathematica.

1.1 Project Repository

The address of this package on GitHub is as follows:

https://github.com/zongpingding/zTeX_bundle

This repository contains the source code of this macro set, the user manual, and some test cases. The current stable version of the macro set was released in September 2025. For the latest version, please switch to the “dev” branch. This manual applies to the current latest “dev” branch.

1.2 Functional Overview

The `zTi $\kern-0.05em k$ Z` package is mainly used for drawing and calculation, supporting calls to external programs such as Python, Mathematica, and gnuplot. It also provides a calling cache mechanism. Although `zTi $\kern-0.05em k$ Z` provides interfaces for these software programs, it does not mean you need to install all of them. Each software interface in `zTi $\kern-0.05em k$ Z` is independent; users only need to install the software corresponding to the functions they need. The functional overview of `zTi $\kern-0.05em k$ Z` is as follows:

- **Drawing:** 2D drawing, 3D drawing;
- **Calculation:** Floating-point calculation, symbolic calculation.

The drawing part is based on: `TikZ`'s 2D drawing part,¹ Python's `matplotlib` library, and WolframScript's drawing functions. The calculation part is based on: `LATEX3`'s `xfp` module, Python's `numpy`, `sympy`, and `scipy` libraries, and WolframScript's calculation functions.

Although the name of this macro set only contains “`TikZ`”, the functions that `ztikZ` can (or aims to) achieve go beyond this.

1.3 Coordinate Alignment

All drawing commands provided by `ztikZ` can be used in conjunction with `TikZ` commands; that is, they can be used within the same `tikzpicture` environment. `ztikZ` performs “alignment” of coordinates when drawing functions: coordinates in `ztikZ` commands are consistent with those in `TikZ` commands or in Geogebra.

Why align coordinates in `ztikZ`? Imagine a scenario: you find the intersection of two function graphs in Geogebra at $P(1, 2)$. First, you draw point P using `TikZ`'s built-in `\filldraw` command; then you draw the same point P using the `\ShowPoint` command in `ztikZ`. The result is that the two P points do not overlap, even though the specified coordinates are both $(1, 2)$.

Therefore, when it is inconvenient to use `ztikZ` to solve for certain special points, you can first find the corresponding point P in software like Geogebra and then directly use the `\ShowPoint` command in `ztikZ` to draw it.

1.4 Cache Mechanism

In addition to providing interfaces for interaction with external programs, `ztikZ` also has a built-in cache system. `ztikZ` automatically caches the results of interactions between `TEX` and external programs and records the Hash value of the calling source code in the `LATEX` document.

If the Hash value of the source code in the `LATEX` document changes, `ztikZ` will re-interact with the external program, generate new results, and cache the new Hash value. If the Hash value does not change, `ztikZ` will directly call the previous cached result. The advantages of the cache system are: we do not have to repeatedly compile unchanged content, directly referencing previous caches, which reduces document compilation time. In actual tests, after results are cached, the time to re-compile the source document is almost the same as the time to directly insert the corresponding number of images.

The `basic`, `python`, `wolfram`, and `gnuplot` libraries in `ztikZ` have all implemented the cache mechanism. The cache mechanism for images generated by the `tikzpicture` environment or the `\tikz` command relies on `TikZ`'s external library (its implementation is notoriously complex; interested users can look it up).

Since `ztikZ` has not yet undergone complete testing, there may be undiscovered bugs. For example, users might encounter problems like the following:

- Outdated Cache Hash value: If the initial Hash value of an environment is “A”, and after you modify the content, the Hash value becomes “B”. However, if you modify the source code back so the Hash value is “A” again, the Hash value is already cached in the `ztikz.hash` file, so the drawing result for this environment will not change

¹Since I haven't decided how to declare the transformation matrix interfaces involved in 3D drawing in `ztikZ`, 3D drawing is currently not provided.

when re-compiled. The cached result called is still the one corresponding to Hash value “B”.

- Conflict with the `indextool` package: You might find a compilation error like `missing \begin{document}...` after enabling the cache library. This issue is related to the indexing function of the `indextool` package. You can comment out the `\makeindex` and `\printindex` commands first, and then uncomment them and generate the index after the image caching is finished.

1.5 Limitations

`ǻTikZ` may provide 3D drawing related interfaces in the future, but if your images require complex calculations or layouts, please use other macro packages or professional drawing software. The `asymptote` package is a good choice.

2 Installation and Usage

2.1 Compatibility

Currently, the `ztikz` package is compatible with Windows, Linux, and macOS. The compatibility with different `TeXLive` versions on each platform is as follows:

Windows : Minimum `TeXLive` version 2023

Linux : Minimum `TeXLive` version 2022

macOS : Minimum `MacTeX` version 2024

The performance of `ztikz` on Windows may not be as good as on Linux/macOS. It is recommended that users use this package on Linux/macOS.

2.2 Environment Configuration

If users need to use the libraries provided by `ztikz` that call external programs, they not only need to configure the document preamble but also need to install the corresponding applications on the system. After the applications are installed, they need to be added to the environment variables so that they can be called from the command line. Finally, when compiling the document, add the `--shell-escape` parameter, like this:

```
pdflatex --shell-escape main.tex
```

On Windows, it is recommended to use the `scoop` package manager to install these software programs, which avoids the hassle of configuring environment variables. The following are points to note during the configuration process for different programs:

2.2.1 gnuplot

On Windows, when using the GUI to install gnuplot, please make sure to check the “Add gnuplot to PATH” option.

2.2.2 Python

If users need to use the functions provided by the `python` library, they need to install Python as well as the `matplotlib`, `sympy`, and `scipy` libraries; the former is used for drawing, and the latter for calculation.

On Windows, due to `TeXLive`’s compilation configuration, ensure that the “.PY” suffix has been removed from the system environment variable `PATHEXT`.

2.2.3 WSTP

If users cannot install Mathematica, the `ztikz` package offers another option – WSTP (Wolfram Symbolic Transfer Protocol). In this case, users only need to install the free WolframScript (Wolfram Engine). For environment configuration methods, please refer to my repository: [WSTP Example](#).

The advantages of this method are obvious: it is free and very fast (about **ten** times faster than calling WolframScript directly). Even if users do not load the `cache` library, the execution speed of all commands in the `wolfram` library is comparable to directly calling the cache.

Currently, the limitations of this method are also obvious: users need to compile the Wolfram engine client themselves and start the service program in advance. This client has only been tested on Linux (contributions for other platforms are welcome).

2.2.4 Wolfram

If users need to use the functions corresponding to the `wolfram` library, they need to install WolframScript or Mathematica. Commands can be chosen to be executed in the cloud, thus avoiding calls to the local Mathematica calculation kernel. Users must first complete the Wolfram account binding on the command line (when calling on the Wolfram Cloud for the first time):

```
> wolframscript -cloud -code 2+2
Wolfram ID: <Account>
Password: <Password>
```

The above command will prompt the user to enter their Wolfram ID and password. After entering the corresponding `<Account>` and `<Password>`, it can be used. However, cloud execution may be slow, and users should decide whether to adopt this solution. The following describes how to install the Wolfram engine locally:

On Linux, software other than Wolfram is easy to install, using the package manager provided by the Linux distribution. Here I provide a method for using Mathematica under Windows in WSL (users can also configure WolframScript without following this method): it's essentially creating a symbolic link from Linux to Windows. Please change the path of WolframScript under Windows in the command according to your actual situation:

```
sudo ln -sf "/mnt/c/Program Files/Wolfram
Research/WolframScript/wolframscript.exe" /usr/bin/wolframscript
```

Please ensure that WolframScript can be called normally from the command line. You can use the following code to test if WolframScript is configured successfully:

```
plotFunction[fun_, xlimits_, ylimits_] := ContourPlot[
  fun, xlimits, ylimits,
  ContourStyle->{
    RGBColor["#00C0A3"],
    Thickness[0.004]
  },
  AspectRatio->((xlimits[[2]]//Abs) +
(xlimits[[3]]//Abs))/((ylimits[[2]]//Abs) + (ylimits[[3]]//Abs)),
  AxesOrigin->{0,0},
  Axes->True,
  Frame->False,
  AxesStyle->Arrowheads[{0, 0.03}],
  AxesLabel->{"x", "y"},
```

```

PlotRange -> Full
]

xlimits = {x, -3, 6};
ylimits = {y, -4, 5};
fp1 = plotFunction[y==Sin[x], xlimits, ylimits];
fp2 = plotFunction[x^2/4 + y^2/3 == 5, {x, -5, 5}, {y, -5, 5}];

figure = Show[fp2, fp1];
(* the saved image format is: '*.wls.pdf' *)
Export["works_well.pdf", figure];

```

Save the above source code as `test.wls`, and then run the following command in the command line:

```
wolframscript -script test.wls
```

If the configuration is successful, a PDF file named `works_well.pdf` will be generated in the current working directory. Otherwise, it indicates that your WolframScript configuration was not successful, and this library cannot be used.

2.2.5 Mathics

Besides choosing WolframScript as the calculation engine, users can also choose Mathics. What is Mathics? An open-source Mathematica Kernel. MathicsScript is a front-end for Mathics with features like automatic command/variable completion and syntax highlighting.

For local installation of Mathics, please refer to: [Installing Mathics3](#). If users have already installed Mathics on Windows and do not want to re-install it in WSL, the method for creating a symbolic link in WSL is the same as for WolframScript. If users installed Mathics via the command “pip install Mathics-omnibus”, the command to create the symbolic link is:

```

sudo ln -sf
"/mnt/c/Users/<name>/AppData/Local/Programs/Python/Python312/Scripts/mathics.exe"
/usr/bin/mathics

```

The `<name>` in the above command needs to be replaced with your own username, and pay attention to the Python version number.

NOTE: Some functions in Mathematica may not be supported by Mathics; please refer to the Mathics documentation.

3 Package Options

 ztikz/library

 New: 2025-05-18

quiet = `<true|false>`.....Initial value: `false`
 library = `<basic|gnuplot|cache|python|wolfram|l3draw>` Initial value: `empty`

The `<library>` option is equivalent to the command `\ztikzloadlib` and is used to specify the list of libraries to be loaded by `zTikZ` when loading the `ztikz` package. The `<quiet>` option controls whether debugging information is displayed in the terminal; setting `<quiet>=true` or loading `<quiet>` alone will suppress most output. A simple configuration example is as follows:

```
\usepackage[library={basic, gnuplot}]{ztikz}
```

Ex. 1

 ztikz/wolfram/engine
 ztikz/wolfram/cloud

 New: 2026-04-29

engine = `<wolfram|wstp|mathics>`.....Initial value: `wolfram`
 cloud = `<true|false>`.....Initial value: `false`

`<engine>` is used to specify the calculation engine for Wolfram code. Currently, Wolfram and Mathics engines are supported; the former is commercial closed-source software, and the latter is open-source. If users have high speed requirements, they can configure the `<wstp>` environment themselves. `<cloud>` is used to specify whether to use Wolfram Cloud for calculation, currently only valid for the `<wolfram>` option; **Note:** Mathics currently does not support cloud computing. A simple configuration example is:

```
\usepackage[
  library = { wolfram },
  wolfram = { engine=wolfram, cloud=true }
]{ztikz}
```

Ex. 2

4 Miscellaneous

`\zTikZ`
`\ztikz`

New: 2025-09-22

These are the logos of the `ztikz` package. Additionally, `ztikz` provides macros related to `hologo`.

```
% \usepackage{hologo}
\zTikZ, \ztikz;\par
\hologo{zTikZ}, \hologo{ztikz}.
```

Ex. 3

```
zTikZ, zTikZ;
zTikZ, zTikZ.
```

`\ztikzMkdir`

New: 2025-05-15

`\ztikzMkdir{⟨path⟩}`

This command is used to create a directory. `⟨path⟩` can be any valid path name, such as `./A/B`.

5 $\text{\textit{zTikZ}}$ Libraries

$\text{\textit{zTikZ}}$ provides various functional libraries that can be loaded using the `\ztikzloadlib` command. Users need to use `\ztexloadlib{<library name>}` to load the corresponding library. The available `<library name>` list in $\text{\textit{zTikZ}}$ is as follows:

- basic
- cache
- gnuplot
- python
- wolfram
- l3draw

All the above libraries are not loaded automatically; users must load them manually. The **basic** library only contains commands for drawing points, lines, axes, and basic polygons. The method to load $\text{\textit{zTikZ}}$ libraries in the preamble is as follows, for example, loading the **cache** and **gnuplot** libraries:

```
\ztikzloadlib{cache, gnuplot}
```

Ex. 4

Note: The script files for a library will only be written to the project folder after the user loads the corresponding library.

5.1 basic Library

The basic library mainly includes commands related to the coordinate system: drawing points, lines, planes, and regular polygons, as well as solving for and drawing intersections. All drawing commands in it are inherited from built-in commands in TikZ; for example, the subsequent `\BarPlot` command is actually a wrapper for the following built-in command:

```
\draw[⟨key-value⟩] plot [ycomb, ⟨other style⟩] file {⟨data⟩};
```

NOTE: For convenience, we use `⟨draw-keyval⟩` to represent the `⟨keyval⟩` options in `\draw[⟨keyval⟩]`, and `⟨node-keyval⟩` to represent the `⟨keyval⟩` options in `\node[⟨keyval⟩]`. Specifically, for the `⟨draw⟩` command, available options include `⟨line width⟩`, `⟨color⟩`, etc.; for details, please refer to the TikZ user manual.

`\ShowPoint`

New: 2025-05-15

```
\ShowPoint[⟨key-value⟩]{⟨point-1⟩; ...; ⟨point-n⟩}
                    [⟨label-1⟩; ...; ⟨label-n⟩][⟨node-keyval⟩]
```

This command is used to draw points. `⟨point-1⟩` to `⟨point-n⟩` are the coordinates of the points, separated by “;”; the coordinate format is (x, y) . `⟨key-value⟩` is used to set the style of the points. The number of `⟨label⟩`s does not need to match the number of `⟨point⟩`s; `⟨label⟩`s are applied sequentially to each point starting from the first.

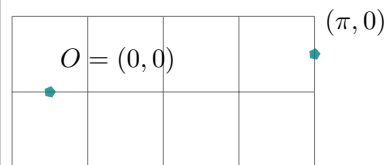
<code>ztikz/point/type</code>	<code>type</code>
<code>ztikz/point/radius</code>	<code>radius</code>
<code>ztikz/point/color</code>	<code>color</code>
<code>ztikz/point/opacity</code>	<code>opacity</code>
<code>ztikz/point/rotate</code>	<code>rotate</code>

<code>type</code>	<code>= ⟨string⟩</code>	Initial value: none
<code>radius</code>	<code>= ⟨length⟩</code>	Initial value: 1pt
<code>color</code>	<code>= ⟨color⟩</code>	Initial value: black
<code>opacity</code>	<code>= ⟨float⟩</code>	Initial value: 1
<code>rotate</code>	<code>= ⟨angle⟩</code>	Initial value: 0

`⟨type⟩` is used to set the style of the marker, `⟨radius⟩` for the radius, `⟨color⟩` for the color, `⟨opacity⟩` for the transparency, and `⟨rotate⟩` for the rotation angle.

```
\begin{tikzpicture}
\draw[gray] (-2, -1) grid (2, 1);
\ShowPoint
  [color=teal, radius=2pt, type=pentagon*, opacity=.8,
rotate=60]
  {(-1.5, 0); (2, .5)}[$O=(0, 0)$; $(\pi, 0)$]
  [above right=3pt and 0em, font=\small]
\end{tikzpicture}
```

Ex. 5



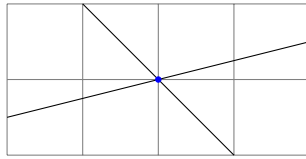
\ShowIntersection

New: 2025-05-15

\ShowIntersection[*<key-val>*]{*<path-1>*; *<path-2>*}{*<number>*}

This command is used to find the intersections of *<path-1>* and *<path-2>*, separated by “;”, and then draws the first *<number>* intersections. *<key-value>* corresponds to the options in the **\ShowPoint** command, i.e., *<ztikz/point>*.

```
\begin{tikzpicture}
\draw[gray] (-2, -1) grid (2, 1);
\draw[name path=line1] (-2, -.5) -- (2, .5);
\draw[name path=line2] (-1, 1) -- (1, -1);
\ShowIntersection[color=blue]{line1; line2}{1}
\end{tikzpicture}
```

Ex. 6**\ShowAxis**

New: 2025-05-15

\ShowAxis[*<key-value>*]{*<start>*; *<end>*}

This command is used to draw an axis. *<start>* and *<end>* represent the starting and ending points of the axis, separated by “;”; the coordinate format is (x,y) . *<key-value>* is an optional parameter for setting the axis style.

ztikz/axis/tickStart	tickStart	= <i><float></i>Initial value: -5
ztikz/axis/tickEnd	tickEnd	= <i><float></i>Initial value: 5
ztikz/axis/axisRotate	axisRotate	= <i><float></i>Initial value: 0
ztikz/axis/mainStep	mainStep	= <i><float></i>Initial value: 1
ztikz/axis/subStep	subStep	= <i><float></i>Initial value: 0.1
ztikz/axis/tickLabelShift	tickLabelShift	= <i><length></i>Initial value: 0pt
ztikz/axis/mainTickLength	mainTickLength	= <i><length></i>Initial value: 4pt
ztikz/axis/subTickLength	subTickLength	= <i><length></i>Initial value: 2pt
ztikz/axis/axisColor	axisColor	= <i><color></i>Initial value: black
ztikz/axis/mainTickColor	mainTickColor	= <i><color></i>Initial value: black
ztikz/axis/subTickColor	subTickColor	= <i><color></i>Initial value: black
ztikz/axis/tickStyle	tickStyle	= <i><below above cross></i>Initial value: none
ztikz/axis/mainTickLabel	mainTickLabel	= <i><string></i>Initial value: \CurrentFp
ztikz/axis/mainTickLabelColor	mainTickLabelColor	= <i><color></i>Initial value: black
ztikz/axis/mainTickLabelPosition	mainTickLabelPosition	= <i><below above cross></i> ... Initial value: below

<mainTickLabel> is mainly used to customize the style of coordinate labels; **\CurrentFp** represents the floating-point value at the current tick. *<tickStyle>* is affected by the *<rotate>* option in the optional parameters of the **tikzpicture** environment.

Note: If the value of the key *<tickStyle>* is not specified when using **\ShowAxis**, no ticks will be drawn.

\CurrentFp

New: 2025-05-31

This command represents the floating-point value at the current tick, which automatically updates at different ticks.

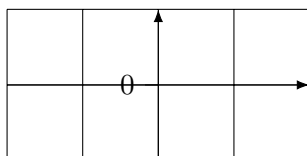
<code>\xAxis</code>	<code>\xAxis[$\langle start \rangle$][$\langle end \rangle$]</code>
New: 2025-05-15	This command comes from <code>\ShowAxis</code> and is used to draw the x -axis; $\langle start \rangle$ and $\langle end \rangle$ are floating-point numbers representing the starting and ending points of the axis.

<code>\yAxis</code>	<code>\yAxis[$\langle start \rangle$][$\langle end \rangle$]</code>
New: 2025-05-15	This command comes from <code>\ShowAxis</code> and is used to draw the y -axis; $\langle start \rangle$ and $\langle end \rangle$ are floating-point numbers representing the starting and ending points of the axis.

```

\begin{tikzpicture}[>=Latex]
\yAxis[-1][1]
\ShowAxis{(-2, 0); (2, 0)}
\draw (-2, -1) grid (2, 1);
\end{tikzpicture}

```

Ex. 7

<code>\ShowGrid</code>	<code>\ShowGrid[$\langle draw-keyval \rangle$]{$\langle start \rangle$; $\langle end \rangle$}</code>
New: 2025-05-15	This command is used to draw grid lines. $\langle start \rangle$ and $\langle end \rangle$ represent the coordinates of the lower-left and upper-right corners of the grid, separated by “;”, in format (x, y) . $\langle key-value \rangle$ is an optional parameter for setting the grid line style.

<code>\Polygon</code>	<code>\Polygon[$\langle key-value \rangle$]{$\langle number \rangle$}</code>
New: 2025-05-15	This command is used to draw regular polygons; $\langle number \rangle$ represents the number of sides, which must be an integer greater than or equal to 3. $\langle key-value \rangle$ is an optional parameter for setting the polygon style.

<code>ztikz/polygon/radius</code>	<code>radius</code>	<code>= $\langle float \rangle$</code>	Initial value: 1
<code>ztikz/polygon/edgeColor</code>	<code>edgeColor</code>	<code>= $\langle color \rangle$</code>	Initial value: black
<code>ztikz/polygon/fillColor</code>	<code>fillColor</code>	<code>= $\langle color \rangle$</code>	Initial value: none
<code>ztikz/polygon/fillOpacity</code>	<code>fillOpacity</code>	<code>= $\langle float \rangle$</code>	Initial value: 0
<code>ztikz/polygon/rotate</code>	<code>rotate</code>	<code>= $\langle float \rangle$</code>	Initial value: 0
<code>ztikz/polygon/shift</code>	<code>shift</code>	<code>= $\langle coordinate \rangle$</code>	Initial value: (0, 0)
<code>ztikz/polygon/marker</code>	<code>marker</code>	<code>= $\langle key-value \rangle$</code>	Initial value: none

$\langle radius \rangle$ represents the radius of the circumscribed circle of this regular polygon, not the radius of the $\langle marker \rangle$. The “()” surrounding $\langle shift \rangle$ cannot be omitted. $\langle marker \rangle$ corresponds to `\ztikz/point`; please see figure (3) for marker settings.

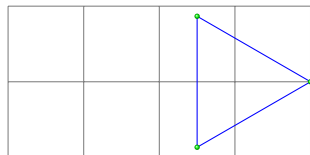
```

\begin{tikzpicture}
\ShowGrid[gray, thin]{(-2, -1); (2, 1)}

```

Ex. 8

```
\Polygon[
  edgeColor=blue, shift={(1, 0)},
  marker={type=ball, color=green}
]{3}
\end{tikzpicture}
```

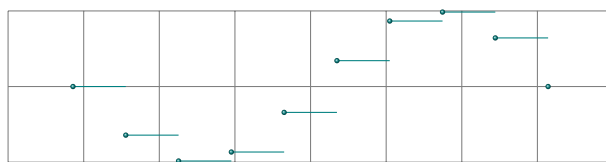


\StairsPlot
New: 2025-05-15

```
\StairsPlot[⟨plot option⟩; ⟨jump option⟩][⟨draw-keyval⟩]
[⟨key-value⟩]{⟨file⟩}
```

This command is used to draw stairs plots, with drawing data specified by *⟨file⟩*. *⟨plot option⟩* is used to set the stairs plot style, with possible values: *plot left*, *plot right*, *plot mid*. *⟨jump option⟩* sets the jump style, with possible values: *jump left*, *jump right*, *jump mid*. *⟨key-value⟩* corresponds to *⟨ztikz/point⟩*.

```
\begin{tikzpicture}
\ShowGrid[step=1, color=gray]{(-4, -1); (4, 1)}
\StairsPlot[; jump-left][teal][type=ball,
color=teal]{./sine.data}
\end{tikzpicture}
```

Ex. 9

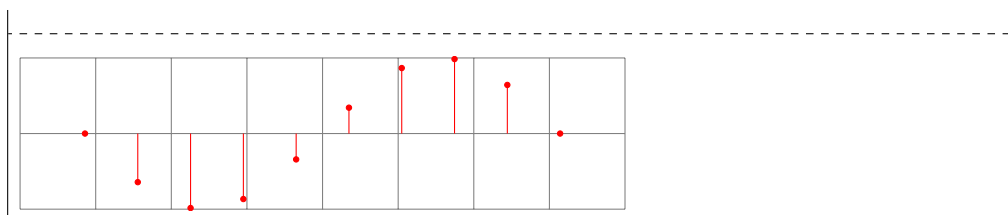
\StemPlot
New: 2025-05-15

```
\StemPlot[⟨direction⟩][⟨draw-keyval⟩]
[⟨key-value⟩]{⟨file⟩}
```

This command is used to draw stem plots, with drawing data specified by *⟨file⟩*. *⟨direction⟩* specifies the direction of the line segments, with possible values: *x*, *y*, *o*, representing perpendicular to the *x*-axis, perpendicular to the *y*-axis, and pointing to the origin, respectively. *⟨key-value⟩* corresponds to *⟨ztikz/point⟩*.

```
\begin{tikzpicture}
\ShowGrid[step=1, color=gray]{(-4, -1); (4, 1)}
\StemPlot[x][red][type=*, color=red]{./sine.data}
\end{tikzpicture}
```

Ex. 10



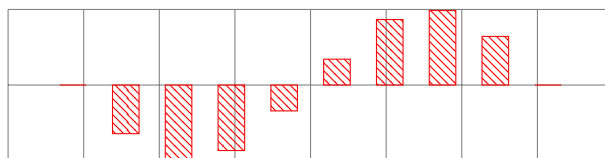
`\BarPlot`

New: 2025-05-15

`\BarPlot[⟨position⟩][⟨draw-keyval⟩`
`[⟨key-value⟩]{⟨file⟩}`

This command is used to draw bar plots, with drawing data specified by `⟨file⟩`. `⟨position⟩` specifies the position and width of each small rectangle, with possible values: `x`, `y`, `xc`, `yc`. `⟨key-value⟩` corresponds to `⟨ztikz/point⟩`.

```
\begin{tikzpicture}
\ShowGrid[step=1, color=gray]{(-4, -1); (4, 1)}
\BarPlot[x][red, pattern=north west lines, pattern
color=red]{./sine.data}
\end{tikzpicture}
```

Ex. 11

5.2 gnuplot Library

It should be noted that the *TikZ* package internally provides commands to directly call the gnuplot program (requires the `-shell-escape` parameter enabled), with the calling format as follows:

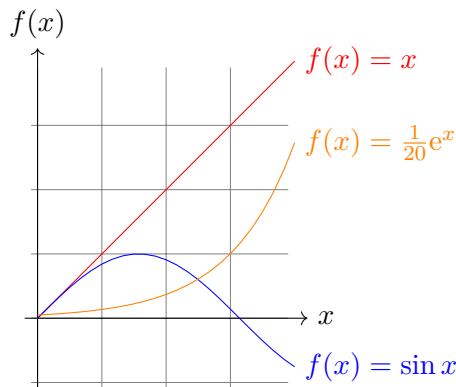
```
\draw[⟨key-value⟩] plot[⟨id⟩] function{⟨function⟩};
```

In the above command, $\langle id \rangle$ is used to distinguish different data files. In the root path of the $\langle file \rangle.tex$ file (let the file name be $\langle file \rangle$), two files will be generated: one is the style file used by gnuplot for drawing, $\langle file \rangle.\langle id \rangle.gnuplot$, and the second is the data file generated by gnuplot, $\langle file \rangle.\langle id \rangle.table$. Possible values for $\langle function \rangle$ can be found in [table \(1\)](#).

TikZ built-in commands also support two other formats: “parametric” and “raw gnuplot”. The first parameter draws parametric equations, and the second parameter uses raw gnuplot drawing commands directly in the document (e.g., “set samples 25; plot sin(x)”). The calling formats are:

```
\draw[⟨key-value⟩] plot [parametric, ⟨id⟩]{⟨function⟩};
\draw[⟨key-value⟩] plot [raw gnuplot, ⟨id⟩]{⟨gnuplot code⟩};
```

```
\begin{tikzpicture}[domain=0:4, scale=.85] Ex. 12
  \draw[very thin,color=gray] (-0.1,-1.1) grid (3.9,3.9);
  \draw[->] (-0.2,0) -- (4.2,0) node[right] {$x$};
  \draw[->] (0,-1.2) -- (0,4.2) node[above] {$f(x)$};
  \draw[color=red] plot[id=x] function{x} node[right] ✓
  {$f(x)=x$};
  \draw[color=blue] plot[id=sin] ✓
  function{sin(x)}node[right]{$f(x)=\sin x$};
  \draw[color=orange] plot[id=exp] function{0.05*exp(x)} ✓
  node[right] {$f(x)=\frac{1}{20}\mathrm{e}^x$};
\end{tikzpicture}
```



For more detailed usage of these native drawing commands in *TikZ*, please

refer to Section 22: Plots of Functions in the official $\text{\textit{TiKZ}}$ documentation.

However, for the uniformity of the $\text{\textit{gnuplot}}$ series of drawing commands, $\text{\textit{ztikz}}$ does not adopt the above method. Instead, it leverages the $\text{\textit{ztool}}$ package and works with predefined drawing scripts to complete the drawing tasks. The drawing logic of the $\text{\textit{gnuplot}}$ library in $\text{\textit{ztikz}}$ is roughly as follows:

- First, modify the predefined script via the $\text{\textit{\ztool_replace_file_line:nnn}}$ function of $\text{\textit{ztool}}$.
- Then call $\text{\textit{gnuplot}}$ to run the modified script using the $\text{\textit{-shell-escape}}$ parameter from the command line.
- Finally, use the command $\text{\textit{\draw[\langle key-value \rangle] plot file [\langle data \rangle]}}$; to call the generated data file from the previous step to complete the drawing.

Users unfamiliar with $\text{\textit{gnuplot}}$ can read this 7-page quick start guide: [gnuplot card](#).

NOTE: After calling this library, the “ $\text{\textit{-shell-escape}}$ ” parameter must be enabled during compilation.

$\text{\textit{ztikz/2dplot/domain}}$	$\text{\textit{domain}} = \langle \text{\textit{float:float; float:float}} \rangle \dots \text{Initial value: } \langle \text{\textit{variable}} \rangle$
$\text{\textit{ztikz/2dplot/style}}$	$\text{\textit{style}} = \langle \text{\textit{draw-keyval}} \rangle \dots \text{Initial value: } \text{\textit{black}}$
$\text{\textit{ztikz/2dplot/marker}}$	$\text{\textit{marker}} = \langle \text{\textit{key-value}} \rangle \dots \text{Initial value: } \text{\textit{empty}}$

$\langle \text{\textit{marker}} \rangle$ with $\langle \text{\textit{key-value}} \rangle$ corresponds to $\langle \text{\textit{ztikz/point}} \rangle$. $\langle \text{\textit{domain}} \rangle$ values are separated by “;”. The meaning of $\langle \text{\textit{domain}} \rangle$ differs across functions: in $\text{\textit{\Plot}}$, it sets the range of the independent variable x ; in $\text{\textit{\ParamPlot}}$ and $\text{\textit{\PolarPlot}}$, it sets the range for parameter t or angle θ in polar coordinates; in $\text{\textit{\ContourPlot}}$, the $\langle \text{\textit{domain}} \rangle$ before and after “;” represent the ranges for x and y , respectively.

$\text{\textit{\PlotPrecise}}$	$\text{\textit{\PlotPrecise}}\{\langle \text{\textit{type}} \rangle\}\{\langle \text{\textit{number}} \rangle\}$
New: 2025-05-15	$\text{\textit{\PlotPrecise}}*\{\langle \text{\textit{type}} \rangle\}\{\langle \text{\textit{number}} \rangle\}$

This command is used to set the precision for a series of **2D** drawing functions in $\text{\textit{gnuplot}}$. Possible values for $\langle \text{\textit{type}} \rangle$ are: “plot, param, polar, contour”, corresponding to the drawing precision of $\text{\textit{\Plot}}$, $\text{\textit{\ParamPlot}}$, $\text{\textit{\PolarPlot}}$, and $\text{\textit{\ContourPlot}}$. Commands with “*” apply to all instances following the drawing command, while those without “*” apply only to the first drawing command that follows.

$\text{\textit{\Plot}}$	$\text{\textit{\Plot}}[\langle \text{\textit{key-value}} \rangle]\{\langle \text{\textit{function}} \rangle\}$
New: 2025-05-15	

This command is used to draw the function $y = y(x)$, where $\langle \text{\textit{function}} \rangle$ is a $\text{\textit{gnuplot}}$ function expression with independent variable “ x ”. $\langle \text{\textit{key-value}} \rangle$ sets the drawing style, corresponding to $\langle \text{\textit{ztikz/2dplot}} \rangle$. $\langle \text{\textit{domain}} \rangle$ defaults to $-5:5$. **Note:** Scatter plots can be achieved simply by setting $\langle \text{\textit{opacity}} \rangle$ to 0.

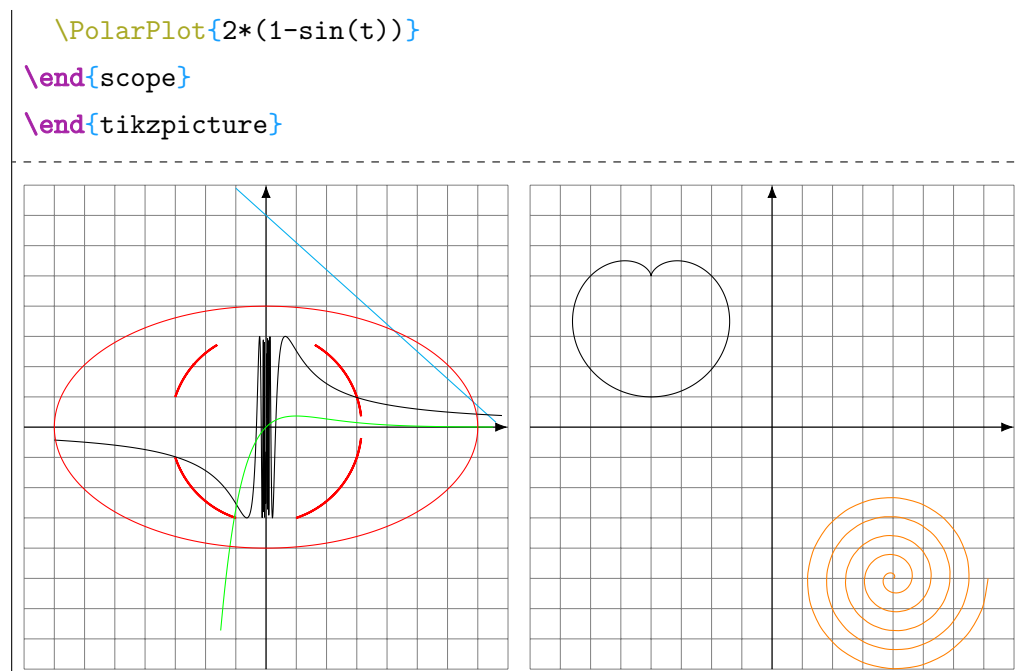
<code>\ContourPlot</code>	<code>\ContourPlot[⟨key-value⟩]{⟨equation⟩}</code>
New: 2025-05-15	This command is used to draw the equation $f(x, y) = c$, where $\langle\text{equation}\rangle$ is a gnuplot equation expression with variables “x”, “y”, and the “$=$” sign is not required in the expression. $\langle\text{key-value}\rangle$ sets the drawing style, corresponding to $\langle\text{ztikz}/2dplot\rangle$. $\langle\text{domain}\rangle$ defaults to “$-5:5;*:*$” (i.e., the range for y is self-adapting). Note: This function can be used to draw vertical segments like $x = c$.
<code>\ParamPlot</code>	<code>\ParamPlot[⟨key-value⟩]{⟨equation⟩}</code>
New: 2025-05-15	This command draws parametric equations $x = x(t)$, $y = y(t)$, where $\langle\text{equation}\rangle$ is a gnuplot equation expression with parameter “t”. $\langle\text{key-value}\rangle$ sets the style, corresponding to $\langle\text{ztikz}/2dplot\rangle$. $\langle\text{domain}\rangle$ defaults to $0:2\pi$.
<code>\PolarPlot</code>	<code>\PolarPlot[⟨key-value⟩]{⟨equation⟩}</code>
New: 2025-05-15	This command draws polar equations $\rho = \rho(t)$, where $\langle\text{equation}\rangle$ is a gnuplot equation expression with parameter “t”. $\langle\text{key-value}\rangle$ sets the style, corresponding to $\langle\text{ztikz}/2dplot\rangle$. $\langle\text{domain}\rangle$ defaults to $0:2\pi$.

```

\begin{tikzpicture}[>=Latex, scale=.4]
\ShowGrid{(-8, -8); (8, 8)}\ShowAxis{(0, -8); (0, 8)} ✓
\ShowAxis{(-8, 0); (8, 0)}
% draw functions/curves
\Plot[domain=-1:7.6, style=cyan] {-.9*x+7}
\ContourPlot[
  domain={-3:pi; -3:exp(1)}, style={red, thick}
]{x**2 + y**2 - 10}
% change plot precise
\PlotPrecise{plot}{1500}
\Plot[domain=-7:7.8] {3*sin(1/x)}
\Plot[domain=-1.5:7.5, style=green] {x*exp(-x)}
\ParamPlot[domain=0:2*pi, style=red] {7*sin(t), 4*cos(t)}
\end{tikzpicture}
\hskip.5em
\begin{tikzpicture}[>=Latex, scale=.4]
\ShowGrid{(-8, -8); (8, 8)}\ShowAxis{(0, -8); (0, 8)} ✓
\ShowAxis{(-8, 0); (8, 0)}
% draw functions/curves
\begin{scope}[xshift=4cm, yshift=-5cm]
  \PolarPlot[domain=0:10*pi, style=orange] {0.1*t}
\end{scope}
\begin{scope}[xshift=-4cm, yshift=5cm]

```

Ex. 13



Reviewing the simple case above: we used the four commands `\Plot`, `\ParamPlot`, `\PolarPlot`, and `\ContourPlot`, and also applied the `\PlotPrecise` command to change the drawing precision of the `\Plot` command.

<code>ztikz/3dplot/domain</code>	<code>domain</code>	<code>= <float:float; float:float>.....Initial value: -5:5; -5:5</code>
<code>ztikz/3dplot/pm3d</code>	<code>pm3d</code>	<code>= <true false>.....Initial value: true</code>
<code>ztikz/3dplot/width</code>	<code>width</code>	<code>= <length>.....Initial value: 0.75\linewidth</code>
<code>ztikz/3dplot/palette</code>	<code>palette</code>	<code>= <string>.....Initial value: rgbformulae 22,13,-31</code>

`<domain>` is used to set the value ranges for independent variables x and y , separated by “;”. `<pm3d>` controls whether to enable surface coloring; if `<pm3d>=false`, only a series of curves for the surface will be drawn. `<width>` sets the width of the image.

<code>\Plotz</code>	<code>\Plotz[<key-value>]{<function>}</code>
New: 2025-05-15	This command draws ordinary 2D explicit functions, where <code><function></code> is a gnuplot function expression. <code><key-value></code> sets the style, corresponding to <code><ztikz/3dplot></code> . Note: This command cannot be used within the <code>\tikzpicture</code> environment.

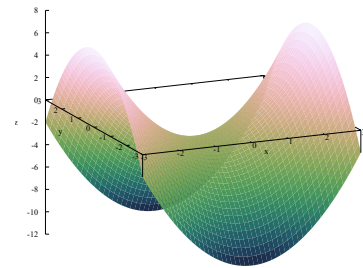
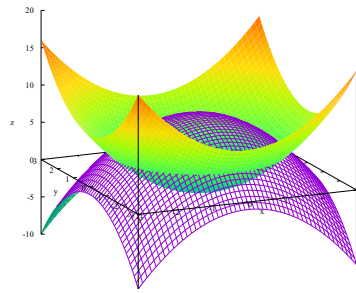
The following case demonstrates the basic usage of the `\Plotz` command, where “ x^2+y^2-2 with `pm3d`” in the first case is a syntax unique to gnuplot; see the gnuplot manual for details.

<pre> \Plotz[pm3d = false, width = .45\linewidth, domain = {-3:3; -3:3}]{x**2+y**2-2 with pm3d, -x**2-y**2+8 with lines} \hskip5em </pre>	Ex. 14
---	--------

```

\Plotz[
  pm3d,
  width = .45\linewidth,
  domain = {-3:3; -3:3},
  palette = {cubehelix start 0 cycles -1. saturation 1}
]{x**2-y**2-2}

```



`\currentTikzIndex`

New: 2025-05-15

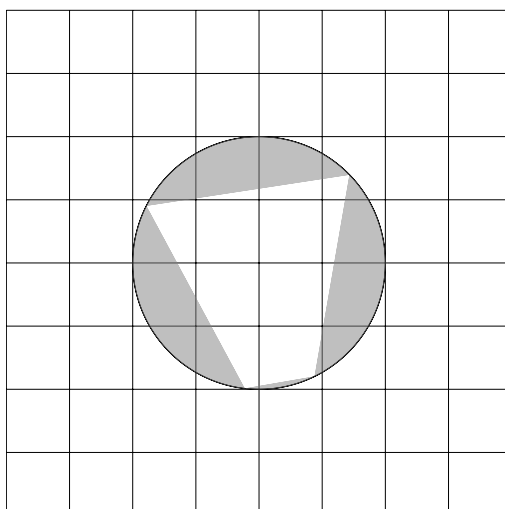
This command represents the index of the current `tikzpicture` environment, returning an integer starting from 1.

`\gnudata` ★ `\gnudata{<index>}`

New: 2025-05-22

This command references the drawing data generated in the current `tikzpicture` environment, returning a (data) file name, starting from 1. `<index>` accepts an integer representing the number of the drawing data in the current environment. Each drawn function will generate a corresponding data file in the corresponding folder, which users can use for subsequent drawing operations.

Note (Supplement to `\gnudata` usage for subsequent area filling): For example, the command `\gnudata{2}` where the parameter “2” indicates this data is the second function drawing data in the current `tikzpicture` environment. Thus, in the first `tikzpicture` environment, its return value might be “./ztikz_output/gnuplot_data/gnu_data_1_2.table”.

Figure 1: `\ContourPlot` Fill Issue

Note: For technical reasons, data generated by the `\ContourPlot` command currently cannot be used for subsequent filling operations. Consider first transforming the implicit function into parametric or polar form, and then exporting the corresponding data. If you force the use of this type of data, you may get an undesirable output like [figure \(1\)](#).

5.3 cache Library

When users load the `cache` library and then compile the document (let its name be $\langle file \rangle$) in the command line, they will see log output like the following:

```
\write18 enabled.
entering extended mode
```

After compilation, a folder named `ztikz_output` will be generated in your project folder, which is created when you first call the `ztikz` macro package. This folder is used to store $\text{\textit{zTikZ}}$'s cache files, including the cached results of $\text{\textit{TikZ}}$'s external library, Python scripts, WolframScript scripts, and gnuplot results.

Now let's talk about the composition of this folder: for example, if a user runs the `\Plot` command, four files as shown in figure (2) will be generated in the `ztikz_output/tikz_data/` directory:

```
ztikz_output
├── gnuplot_data ..... gnuplot cache dir
│   └── gnu_data_1_1.table
├── mma_data ..... WolframScript cache dir
├── python_data ..... Python cache dir
├── scripts ..... gnuplot scripts
│   ├── 3d_plot.gp
│   ├── contour_plot.gp
│   ├── param_plot.gp
│   ├── plot_plot.gp
│   └── polar_plot.gp
├── tikz_data ..... TikZ cache dir
│   ├──  $\langle file \rangle$ -figure0.dpth
│   ├──  $\langle file \rangle$ -figure0.log
│   ├──  $\langle file \rangle$ -figure0.md5
│   ├──  $\langle file \rangle$ -figure0.pdf
│   └──  $\langle file \rangle$ -figure0.run.xml
└── ztikz.hash ..... Hash records
```

Figure 2: Schematic of the $\text{\textit{zTikZ}}$ Cache Directory Structure

$\langle file \rangle$ -figure0.pdf in `tikz_data` is the cached PDF file of the `tikzpicture` environment. At this time, the following content can be seen in the corresponding $\langle file \rangle$.md5 file:

```
\def \tikzexternallastkey {AE7F2539E81C96848ADCCEE3994993D1}%
```

The above command saves the Hash value of the code in this `tikzpicture` environment. When we change the code in the environment, this Hash value will change, causing $\text{\textit{TikZ}}$ to run the environment again and re-generate the image. This is a rough description of the cache function provided by $\text{\textit{TikZ}}$'s external library. The Cache mechanism in $\text{\textit{zTikZ}}$ is very similar to this principle.

5.3.1 User Interface

`\ztikzHashClean`

New: 2025-05-15

This command takes no parameters and is used to clear all previously cached Hash values.

`\ztikzHashCurrent`

New: 2025-05-15

`\ztikzHashCurrent*`
`\ztikzHashCurrent[⟨separator⟩]`

This command is mainly used for debugging cache-related code, often in conjunction with the `\ztikzForceToSkip` command. `\ztikzHashCurrent*` outputs the most recent Hash calculation result. `\ztikzHashCurrent[⟨separator⟩]` outputs all cached Hash values up to now, separated by `⟨separator⟩` in the PDF; `⟨separator⟩` defaults to “,”.

`\ztikzCachedHash *`

New: 2025-05-29

`\ztikzCachedHash[⟨keyval⟩]`

This command is used to output the currently cached Hash value, with a wider range of applications than the `\ztikzHashCurrent` command.

`ztikz/cache/hash/index`

`ztikz/cache/hash/file`

`ztikz/cache/hash/label`

`index = ⟨integer⟩.....Initial value: -1`
`file = ⟨filename⟩.....Initial value: ztikz_output/ztikz.hash`
`label = ⟨string⟩.....Initial value: \g_zcache_latest_cache_label_tl`

By default, `⟨label⟩` is the latest cache label; `⟨file⟩` is the cache file for Hash values; `⟨index⟩` specifies the index of the Hash value cached by that `⟨label⟩`, defaulting to -1 (the latest Hash value).

`\ztikzForceToSkip`

New: 2025-05-15

This command forces skipping the (re-)running of the first environment or command with a cache mechanism that follows it, even if the Hash for that environment or command has changed. Subsequent referenced results from `\wolframResult` or `\wolframOutputFile` commands will be affected by this command.

Note: When this command is applied, new Hash values will not be cached. Currently, this command only works for commands and environments in the `python` and `wolfram` libraries; it will delete scripts and results related to the **new** Hash.

`\ztikzForceToRun`

New: 2025-05-21

This command forces the running of the first environment or command with a cache mechanism that follows it, even if its Hash has not changed.

Note: Currently, this command only works for commands and environments in the `python` and `wolfram` libraries. Even if the Hash value of subsequent commands or environments changes, this new Hash value will not be cached.

5.3.2 Programming Interface

The `cache` library of `ztikz` provides a series of programming interfaces that users can use to write commands or environments related to calling external programs. These new commands or environments will support the cache mechanism. The programming interfaces and some other related commands currently provided in the `cache` library are:

`\g_ztikz_file_hash_seq`

New: 2025-05-30

This sequence (variable) stores all currently cached Hash values, where each item is the Hash value corresponding to a command or environment with a cache mechanism.

Note: This command is set by the subsequent `\__zcache_hash_extract_all:nN` command.

`\g_ztikz_hash_nochg_run_bool`

`\g_ztikz_hashchg_norun_bool`

New: 2025-05-30

These two boolean values control the behavior of the `\ztikz_if_run_again:nnnTF` command. When the former is “true”, Hash verification for the `\ztikz_if_run_again:nnnTF` command will be forced to skip and the `<true code>` will be run directly. When the latter is “true”, Hash verification will be forced to skip and the `<false code>` will be run directly.

Note: Both boolean values default to “false”; it is an error for both to be “true” at the same time.

`\xsim_file_write_start:nn`

`\xsim_file_write_start:ne`

New: 2025-05-30

`\xsim_file_write_start:nn <bool>{<file>}`

This command from the `xsimverb` package is used to transcribe environment content into the `<file>` file. `<bool>` is a boolean; if the environment needs to accept parameters, set `<bool>` to “`\c_true_bool`”, otherwise to “`\c_false_bool`”.

Note: This command must be used in conjunction with the `\xsim_file_write_stop:` command, otherwise an error will occur.

`\xsim_file_write_stop:`

New: 2025-05-30

This command ends the transcription of environment content and must be used with the `\xsim_file_write_start:nn` command.

`\ztikz_if_run_again:nnnTF`

`\ztikz_if_run_again:nenTF`

New: 2025-05-30

`\ztikz_if_run_again:nnnTF`
`{<bool>}{<file/hash>}{<label>}`
`{<true code>}{<false code>}`

`<bool>` controls the type of the second parameter `<file/hash>`. If `<bool>` is “`\c_true_bool`”, `<file/hash>` should be a file name; otherwise, it should be a Hash value. `<label>` is the cache label corresponding to that Hash or file Hash. When the Hash value for the environment does not exist, changes, or follows the `\ztikzForceToRun` command, `<true code>` will be executed. When the Hash value exists or follows the `\ztikzForceToSkip` command, `<false code>` will be executed.

```
\ztikz_term_info:n \ztikz_term_info:n {message}
```

```
\ztikz_term_info:e
```

This command is similar to the `\typeout` command in T_EX, used to output information *message* to the terminal.

New: 2025-05-31

5.3.3 Private Interface

This section describes the private commands and variables in the `cache` library, which may be necessary when users define some commands or environments. As these commands have not yet been fully organized, they may undergo significant changes in the future, and users should use them with caution.

<code>\l_ztikz_current_hash_tl</code>	This variable returns the latest Hash value.
---------------------------------------	--

New: 2025-05-30

<code>_zcache_hash_add:nn</code>	<code>_zcache_hash_add:nn {\label}{\hash}</code>
-----------------------------------	---

New: 2025-05-30

This command adds `\hash` to the cache label corresponding to `\label`.

<code>_zcache_hash_extract_all:nN</code>	<code>_zcache_hash_extract_all:nN {\file}\seq</code>
---	---

New: 2025-05-30

This command extracts all Hash values from the file `\file` and saves them into the variable `\seq`, where each item corresponds to a Hash value.

<code>_zcache_hash_extract_by_label:nnn</code>	<code>_zcache_hash_extract_by_label:nnn {\file}{\label}{\index}</code>
---	---

New: 2025-05-30

This command extracts the Hash value with label `\label` and index `\index` from file `\file`. It has no return value but sets `\l_ztikz_current_hash_tl` to the extracted Hash value.

5.3.4 Programming Interface Case Study

Below is a usage example of the above commands, creating a Mermaid drawing environment that supports the cache mechanism (users need to install the “mermaid-cli” tool):

Ex. 15

```

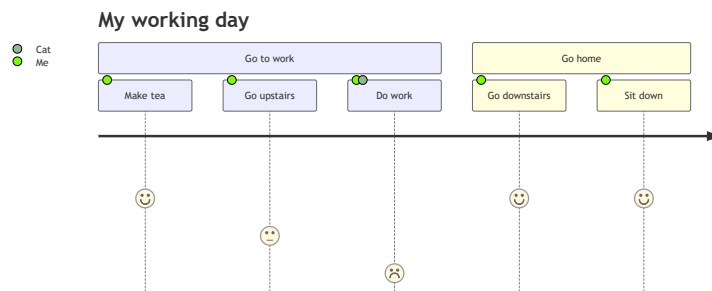
\ExplSyntaxOn\makeatletter
% 1. environment declaration
\tl_new:N \g__ztikz_mmd_path_tl
\tl_gset:Nn \g__ztikz_mmd_path_tl {ztikz_output/mmd_data}
\ztool_shell_mkdir:e { \g__ztikz_mmd_path_tl }
\NewDocumentEnvironment{mermaid}{m }
{
  \exp_args:Ne \xsim_file_write_start:nn {\c_true_bool}{
    \g__ztikz_mmd_path_tl/t@mp.mmd}
  }{
    \xsim_file_write_stop:
    \ztikz_if_run_again:nenTF {\c_true_bool}{t@mp.mmd}{#1}
    {
      \ztool_shell_escape:e
      {
        mmdc
        \space-i\space \g__ztikz_mmd_path_tl/t@mp.mmd
        \space-o\space \g__ztikz_mmd_path_tl/t@mp.pdf
        \space-f
      }
      \ztikz_term_info:e { Mermaid~running~on~file:'
\l__ztikz_current_hash_tl.mmd'~... }
    }{
      \ztikz_term_info:e { Use~the~cached~Mermaid~result:'
\l__ztikz_current_hash_tl.pdf'~... }
    }
    \edef\t@mp@file{\g__ztikz_mmd_path_tl/\l__ztikz_current_hash_tl}
    \ztool_shell_mv:ee
    {\g__ztikz_mmd_path_tl/t@mp.mmd}
    {\t@mp@file.mmd}
    \ztool_shell_mv:ee
    {\g__ztikz_mmd_path_tl/t@mp.pdf}
  }
}

```

```

    {\t@mp@file.pdf}
\edef\mmdOutputFile{\t@mp@file.pdf}
}
\makeatother\ExplSyntaxOff
% 2. environment usage
\begin{mermaid}{mmd-I}
journey
  title My working day
  section Go to work
    Make tea: 5: Me
    Go upstairs: 3: Me
    Do work: 1: Me, Cat
  section Go home
    Go downstairs: 5: Me
    Sit down: 5: Me
\end{mermaid}
\begin{center}
  \includegraphics[width=.7\linewidth]{\mmdOutputFile}
\end{center}

```



5.4 python Library

The `python` library is mainly used for interacting with Python, with usage similar to the `gnuplot` library. It primarily provides image drawing and calculation interfaces, where calculation interfaces include numerical and symbolic calculations.

Apart from the Python drawing functions provided by $\text{\texttt{zTikZ}}$, we need to highlight the floating-point calculation functions: when calling this library, $\text{\texttt{zTikZ}}$ imports the Python packages `numpy`, `sympy`, and `scipy` by default. Furthermore, when using functions in `numpy`, users no longer need to add prefixes; for example, when solving for `sin(2.345)`, use `\py{sin(2.345)}` directly instead of `\py{np.sin(2.345)}`. The same applies to functions in other Python libraries.

NOTE: After calling this library, the “`-shell-escape`” parameter must be enabled during compilation.

 $\text{\texttt{\py}}$

New: 2025-05-15

`\py[ $\langle\text{raw}\rangle$ | $\langle\text{str}\rangle$ ]{ $\langle\text{code}\rangle$ }`

This command calls Python for floating-point calculation, with $\langle\text{code}\rangle$ being valid Python code. These results are not cached, meaning Python re-calculates them each time the document is compiled. Users can nest the `\py` command into their own defined macros.

Note: $\langle\text{raw}\rangle$ tokenizes the returned results according to original $\text{\texttt{T\TeX}}$ cat-codes, while $\langle\text{str}\rangle$ treats the results as strings.

```
\newcommand{\pypow}[1]{\py{#1}}
\newcommand{\pyreverse}[1]{\py{'#1'[::-1]}}
\newcommand{\pyuppercase}[1]{\py{'#1'.upper()}}
\begin{itemize}
  \item Power Calculation:  $2^{10} = \text{\texttt{\pypow{2**10}}}$ 
  \item Reverse a string using Python:  $\text{\texttt{\pyreverse{Hello-Lat\TeX}}}$ 
  \item Uppercase a string:  $\text{\texttt{\pyuppercase{hello-latex}}}$ 
  \item Modulus:  $102 = \text{\texttt{\py{mod(102, 8)}}} \bmod 8$ 
  \item Return string Options:  $\text{\texttt{\py[str]{'\$'+str(2**10)+'\$'}}}$ 
\end{itemize}
```

Ex. 16

-
- Power Calculation: $2^{10} = 1024$
 - Reverse a string using Python: XeTaL-olleH
 - Uppercase a string: HELLO-LATEX
 - Modulus: $102 = 6 \bmod 8$
 - Return string Options: \$\$1024\$\$

 $\text{\textit{\sympy}}$

New: 2025-05-29

 $\text{\textit{\sympy}}\{\langle label \rangle\}\{\langle expression \rangle\}$

This command is used to call Python's `sympy` library for symbolic calculations, and $\text{\textit{zTikZ}}$ provides a cache mechanism for it. $\langle label \rangle$ cannot contain ":" and is used to specify the cache label, which must be unique in the current document. $\langle expression \rangle$ is the symbolic expression. A series of symbolic variables are predefined in the `python` library, including `x`, `y`, `z`, `u`, `v`, `t`, which do not need re-declaration.

Note: By default, results may include characters like " $\wedge$ ", " $_$ " that can only be used in math mode, so users should place this command within a mathematical environment.

Ex. 17

```

\[\int x^8 + \cos(7x) + 6t\,, \mathrm{d}x
= \sympy{integral}{integrate( x**8 + cos(7*x) + 6*t, x )}
\]
\[\mathrm{eig}(\begin{bmatrix}1 & 2\\2 & 2\end{bmatrix})
= \sympy{matrix}{Matrix([[1, 2], [2, 2]]).eigenvals()}
\]
```

$$\int x^8 + \cos(7x) + 6t \, dx = 6tx + \frac{x^9}{9} + \frac{\sin(7x)}{7}$$

$$\mathrm{eig}\left(\begin{bmatrix}1 & 2 \\ 2 & 2\end{bmatrix}\right) = \left\{\frac{3}{2} - \frac{\sqrt{17}}{2} : 1, \frac{3}{2} + \frac{\sqrt{17}}{2} : 1\right\}$$

 $\text{\textit{pyfig}}$

Updated: 2025-05-29

 $\text{\textit{\begin{pyfig}}}\{\langle label \rangle\}\{\langle output \text{ file} \rangle\}$
 $\quad \langle plot \text{ code} \rangle$
 $\text{\textit{\end{pyfig}}}$

This environment calls Python for drawing and returns no results. It has a cache mechanism. $\langle label \rangle$ cannot contain ":" and must be unique. $\langle output \text{ file} \rangle$ specifies the name of the output file for the $\langle plot \text{ code} \rangle$, without the path but with the extension.

Note: It is recommended to use different $\langle output \text{ file} \rangle$ values for different `pyfig` environments. Users do not need to add `plt.savefig()`; $\text{\textit{zTikZ}}$ handles this. Indentation is preserved starting from the beginning of the line, so avoid extra leading spaces. Ensure $\langle output \text{ file} \rangle$ matches the file name in the environment code.

 $\text{\textit{\pyfigOutputFile}} \star$

New: 2025-04-21

This command returns the name of the file output by the `pyfig` environment, which can be imported using commands like `\input` or `\includegraphics`.

Ex. 18

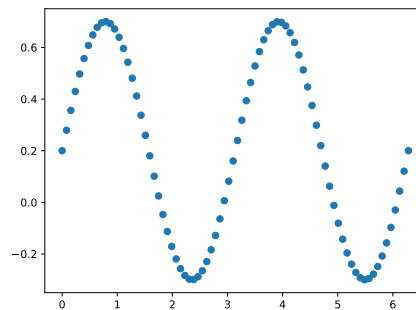
```

\begin{pyfig}{sinGraph}{sin_graph.pdf}
import matplotlib
```

```

matplotlib.use('Agg')
from matplotlib import pyplot as plt
import numpy as np
x = np.linspace(0, 2*np.pi, num = 80)
y = np.sin(x)*np.cos(x)+.2
plt.plot(x, y, 'o')
\end{pyfig}
\begin{center}
  \includegraphics[width=.5\linewidth]{\pyfigOutputFile}
\end{center}

```



`pycode`

New: 2025-05-21

```

\begin{pycode}{\label}{\langle output file \rangle}
  \langle any python code \rangle
\end{pycode}

```

This environment calls Python to execute $\langle \text{any python code} \rangle$ and returns no results. It has a cache mechanism. $\langle \text{label} \rangle$ cannot contain “:” and must be unique. $\langle \text{output file} \rangle$ specifies the name for the output file without the path but with the extension. $\text{\textit{zTikZ}}$ automatically calls Python to execute the file, and results are saved in `\pycodeOutputFile`.

Note: It is recommended to use different $\langle \text{output file} \rangle$ values for different `pycode` environments. Indentation is preserved, so avoid excessive indentation. Ensure $\langle \text{output file} \rangle$ matches the file name in the environment code.

`\pycodeOutputFile` ★

New: 2025-04-21

Returns the name of the file output by the `pycode` environment, which can be imported using `\input` or `\includegraphics`.

Below is a simple usage example for the `pycode` environment; for the content of `table.py.txt`, see [section \(6.3\)](#).

```

\input{./table.py.txt}
\begin{center}
  \input{\pycodeOutputFile}

```

Ex. 19

`\end{center}`

number/function	sin	cos	tan
1	0.8415	0.5403	1.5574
2	0.9093	−0.4161	−2.185
3	0.1411	−0.99	−0.1425
4	−0.7568	−0.6536	1.1578
5	−0.9589	0.2837	−3.3805
6	−0.2794	0.9602	−0.291
7	0.657	0.7539	0.8714
8	0.9894	−0.1455	−6.7997
9	0.4121	−0.9111	−0.4523
10	−0.544	−0.8391	0.6484
11	−1.0	0.0044	−225.9508
12	−0.5366	0.8439	−0.6359
13	0.4202	0.9074	0.463
14	0.9906	0.1367	7.2446
15	0.6503	−0.7597	−0.856

5.5 wolfram Library

The `wolfram` library in `zTikZ` can be seen as a new implementation of the original `latexalpha2` package, addressing its various limitations. Currently, the `wolfram` library has implemented all commands from `latexalpha2` except for `\wolframanimation`, with significant improvements in compatibility, ease of use, and extensibility². For example, it can be used on Windows, Linux, and macOS, and environment source code supports special characters like “\, #, \$, _, ^, &”. Here are some points to note:

- Note the comment style in WolframScript: use “`(* something *)`” rather than “`(* something*)`”; comments should not be adjacent to “`*`” to avoid parsing errors.
- Due to WolframScript limitations, scripts must have the “`.wls`” extension to be recognized and executed.
- For all subsequent commands or environments supporting caching in this section, the cache label `\label` can only use “**numbers, letters, hyphens (-)**”.

NOTE: After calling this library, the “`-shell-escape`” parameter must be enabled during compilation.

<code>\wolframResult</code>
New: 2025-05-15
Updated: 2026-05-07

`\wolframResult[separator]`
`\wolframResult*[index]`

This command references results from the previous WolframScript calculation. `\wolframResult[separator]` references all results separated by `separator`. `\wolframResult*[index]` references only a specific result, where `index` is an integer, defaulting to 1.

NOTE: Because `\wolframResult` partitions results based on “`@12`”, ensure the catcode of “`@`” is 12, otherwise parsing errors may occur.

<code>\wolframOutputPath</code> ★
New: 2026-05-08

This command stores the folder where WolframScript running results (files) are located; it is often used with the `wolframCode` environment to reference image or text results. This command is more general than the subsequent `\wolframResult` and `\wolframOutputFile` commands.

<code>\wolframOutputFile</code> ★
New: 2025-05-15
Updated: 2026-04-30

Returns the filename corresponding to the previous WolframScript run; very convenient for referencing image results. It is more flexible than `\wolframResult` as the former calls the text file while the latter returns only the filename produced by the last WolframScript call.

²2026-04-29: Speed has also significantly increased with the new `wstp` engine.

`\wolfram`

New: 2025-05-29

`\wolfram{<label>}{<code>}`
`\wolfram*{<label>}{<code>}`

Calls WolframScript for calculation with a cache mechanism. `<label>` cannot contain “:” and must be unique. `<code>` is valid WolframScript code. Results are $\text{\LaTeX}$ code by default; commands with “*” return ordinary strings.

```
\wolfram{wolframLaplace}{LaplaceTransform[t^4 Sin[3*t], t, s]}
\[\mathcal{L}(t^4\sin(3t)) = \wolframResult
\]
```

$$\mathcal{L}(t^4 \sin(3t)) = \frac{72(5s^4 - 90s^2 + 81)}{(s^2 + 9)^5}$$

`\wolframTex`

New: 2025-05-29

`\wolframTex{<label>}{<Tex code>}`

Similar to `\wolfram`, but it converts all content in `<Tex code>` into Mathematica/Mathics code, returning results as $\text{\LaTeX}$ code. `<label>` cannot contain “:” and must be unique.

NOTE: Due to the complex and special implementation, neither `<Tex code>` nor `<label>` can contain the “\$” symbol, otherwise parsing errors will occur; users should not rely too much on this command, as it may consume significant time or even cause compilation errors in some cases.

```
\wolframTex{wolframTexInt}{\int_a^b \sin(x) \, dx}
\[\int_a^b \sin(x) \, dx = \wolframResult
\]
```

$$\int_a^b \sin(x) \, dx = \cos(a) - \cos(b)$$

`\wolframTable`

New: 2025-05-29

`\wolframTable{<label>}{<code>}`
`\wolframTable*{<label>}[<key-value>]{<code>}`

Calls the Wolfram engine to generate a table with a cache mechanism. `<label>` cannot contain “:” and must be unique. Commands without “*” do not output the table in the PDF but set `\wolframTablePData` and `\wolframTableFData` variables. Commands with “*” output the table with styles specified by `<key-value>`. `<code>` is valid WolframScript/Mathics code.

```
\wolframTable*{wolframTable}[
format=cccc, hdbt-rule,
header={x$ & x^2$ & x^3$ & x^4$},
cell-cmd={\textcolor{red}{(1)}}]
\]
```

```
] {Table[{i, i^2, i^3, i^4}, {i, 6}]}
\SetTblrOuter{expand=\wolframTableFData}
\hskip6em
\begin{tblr}
{
  colspec = {cccc},
  rowspec = {
    | [2pt,green7]Q| [2pt, teal7]Q| [green7]Q| [green6]
    Q| [green5]Q| [green4]Q| [green3]Q| [3pt,teal7]
  }
} \wolframTableFData
\end{tblr}
```

x	x^2	x^3	x^4
(1)	(1)	(1)	(1)
(2)	(4)	(8)	(16)
(3)	(9)	(27)	(81)
(4)	(16)	(64)	(256)
(5)	(25)	(125)	(625)
(6)	(36)	(216)	(1296)

x	x^2	x^3	x^4
(1)	(1)	(1)	(1)
(2)	(4)	(8)	(16)
(3)	(9)	(27)	(81)
(4)	(16)	(64)	(256)
(5)	(25)	(125)	(625)
(6)	(36)	(216)	(1296)

ztikz/wolfram/table/format	format	= $\langle$ column format $\rangle$	Initial value: $\ast\{12\}\{1\}$
ztikz/wolfram/table/header	header	= $\langle$ header $\rangle$	Initial value: empty
ztikz/wolfram/table/hdbt-rule	hdbt-rule	= $\langle$ false true $\rangle$	Initial value: false
ztikz/wolfram/table/cell-cmd	cell-cmd	= $\langle$ cell command $\rangle$	Initial value: #1

$\langle$ format $\rangle$ sets column formatting; $\langle$ header $\rangle$ sets the table header for the first row; $\langle$ hdbt-rule $\rangle$ (header bottom rule) specifies whether to display a horizontal line below the header (only for $\backslash$ wolframTable $\ast$); $\langle$ cell-cmd $\rangle$ accepts one parameter and is applied to each cell in $\backslash$ wolframTablePData (excluding the header), using “#1” for the content.

<code>\wolframTablePData</code> ★	These commands represent the results of the most recent <code>\wolframTable</code> calculation without commands like <code>\hline</code> or <code>\midrule</code> . <code>\wolframTablePData</code> (Part Data) excludes the header, while <code>\wolframTableFData</code> (Full Data) includes it. They can be passed to table formatting packages like <code>tabularray</code> . Note: <code>\cell-cmd</code> in <code>\wolframTable</code> should not be mixed with <code>\cells/cmd</code> in <code>tabularray</code> .
<code>\wolframTableFData</code> ★	
New: 2025-05-18	

`\wolframSolve` `\wolframSolve{⟨label⟩}[⟨key-value⟩]{⟨equation⟩}`
`\wolframSolve*{⟨label⟩}{⟨full code⟩}`

New: 2025-05-29

Calls WolframScript to solve equations with a cache mechanism. `⟨label⟩` cannot contain “:” and must be unique. `⟨equation⟩` is the equation expression. `⟨key-value⟩` sets independent variables and domains. `⟨full code⟩` is the complete equation expression including variables and domains.

`ztikz/wolfram/solve/domain` `domain = ⟨domain⟩.....Initial value: empty`
`ztikz/wolfram/solve/var` `var = ⟨variable⟩.....Initial value: empty`

`⟨domain⟩` sets the range for solving (e.g., `⟨domain⟩=Integers`); `⟨var⟩` sets the independent variables (e.g., `⟨var⟩=x`).

```
\wolframSolve{wolframSolve-I}[var={x, y}]{a x + y == 8 && b x + c y == 1}
```

```
\begin{align}
```

```
& \wolframResult \\\
```

```
& \wolframResult[||] \\\
```

```
& \wolframResult* \\\
```

```
& \wolframResult*[3-1]
```

```
\end{align}
```

```
\wolframSolve{wolframSolve-II}
```

```
[var={x, y}, domain=Integers]
```

```
{x^2 + 2 y^3 == 3681 && x > 0 && y > 0}
```

```
\begin{align}
```

```
\wolframResult
```

```
\end{align}
```

$$x = \frac{9}{a+b}, y = -\frac{a-8b}{a+b} \quad (5.1)$$

$$x = \frac{9}{a+b} || y = -\frac{a-8b}{a+b} \quad (5.2)$$

$$x = \frac{9}{a+b} \quad (5.3)$$

$$y = -\frac{a-8b}{a+b} \quad (5.4)$$

$$x = 15, y = 12, x = 41, y = 10, x = 57, y = 6 \quad (5.5)$$

`\wolframDSolve` `\wolframDSolve{<label>}[<key-value>]{<equation>}`
 `\wolframDSolve*{<label>}{<full code>}`

New: 2025-05-29

Calls WolframScript to solve differential equations with a cache mechanism. `<label>` cannot contain “:” and must be unique. `<equation>` is the equation expression. `<key-value>` sets variables; `<full code>` is the complete expression.

ztikz/wolfram/dsolve/depend `depend` = `<dependent variable>.....Initial value: y[x]`
 ztikz/wolfram/dsolve/independ `independ` = `<independent variable>.....Initial value: x`

`<depend>` specifies the dependent variable (e.g., `<depend>=y[x]`); `<independ>` specifies the independent variable (e.g., `<independ>=x`).

Ex. 24

```

\wolframDSolve{wolframDSolve-I}
{y'[x] + y[x] == a*Sin[x], y[0] == 1}
\begin{align}
&\&\wolframResult
\end{align}
\wolframDSolve{wolframDSolve-II}
[depend={y[x], z[x]}]
{y'[x] == Exp[z[x]] + 1, z'[x] == y[x] - x}
\begin{align}\left\{\begin{aligned}
&\&\wolframResult[\&]
\end{aligned}\right.\end{align}
\end{aligned}\right.\end{align}

```

(5.6)

$$y(x) = -\frac{1}{2}e^{-x}(-ae^x \sin(x) + ae^x \cos(x) - a - 2)$$

(5.7)

$$\begin{cases} z(x) = \log\left(c_1 \tan^2\left(\frac{1}{2}\left(\sqrt{2}\sqrt{c_1}x + 2\sqrt{2}\sqrt{c_1}c_2\right)\right) + c_1\right) \\ y(x) = x + \sqrt{2}\sqrt{c_1} \tan\left(\frac{1}{2}\left(\sqrt{2}\sqrt{c_1}x + 2\sqrt{2}\sqrt{c_1}c_2\right)\right) \end{cases}$$

`wolframGraphics` `\begin{wolframGraphics}{<label>}[<spec>]`
 `<plot code>`

`\end{wolframGraphics}`

New: 2025-05-29

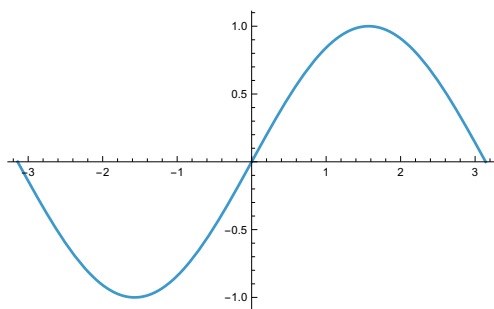
Environment for drawing with WolframScript, with a cache mechanism. `<label>` cannot contain “:” and must be unique. `<spec>` sets layout parameters; if empty, no result is returned and the file can be called via `\wolframOutputFile`. If `<spec>` is not empty, the return value is `\includegraphics[<spec>]{<path>/<HASH>.pdf}`.

NOTE: The final image name in `<plot code>` must be “FIGURE”, otherwise an error will occur.

```

\begin{wolframGraphics}{wolframSinGraph}
    FIGURE=Plot[Sin[x], {x, -Pi, Pi}]
\end{wolframGraphics}
% include graphics
\includegraphics[width=.5\linewidth]{\wolframOutputFile}

```

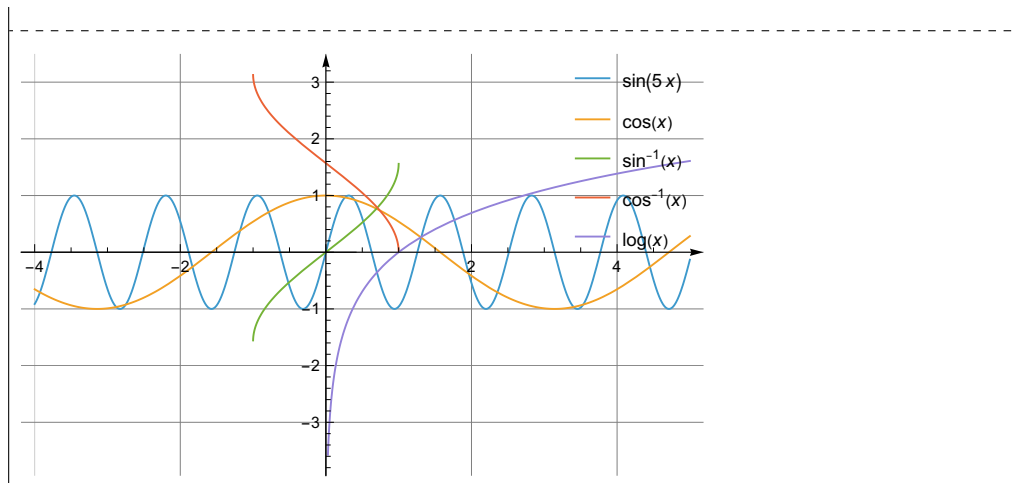
Ex. 25

```

\begin{wolframGraphics}{fig-1}[width=.7\linewidth]
fp = Plot[{
    Sin[5 x],
    Cos[x],
    ArcSin[x],
    ArcCos[x],
    Log[x]
},
{x, -4, 5},
PlotLegends -> Placed["Expressions", {Right, Top}],
PlotStyle -> AbsoluteThickness[1],
AxesStyle -> Arrowheads[0.02],
GridLines -> Automatic
];
FIGURE = Show[fp];
\end{wolframGraphics}

```

Ex. 26



```
wolframCode \begin{wolframCode}{\label}
              \langle wolfram code \rangle
New: 2026-05-08 \end{wolframCode}
```

This environment can run arbitrary Wolfram code and has a cache mechanism. If users need to export calculation results, they can use the “`scriptPath`” variable provided by this `wolfram` library, which represents the cache folder for the Wolfram code. $\langle label \rangle$ cannot contain “.” and must be unique within the current document.

NOTE: The cache mechanism of this environment has a flaw: if the $\langle label \rangle$ has already cached results for two hashes “X, Y”, and the user changes the code such that the new Hash happens to be “X”, the compilation will only return the result corresponding to “Y”. Users can add the `\ztikzForceToRun` command before this environment to refresh the cache.

```
\begin{wolframCode}{wolfram-code-1} Ex. 27
num = 2^110;
Export[scriptPath <> "wolfram-code_text.txt", num]
\end{wolframCode}
% load result
\[ 2^{\{110\}} = \input{\wolframOutputPath/wolfram-code_text.txt} \]
```

$$2^{110} = 1298074214633706907132624082305024$$

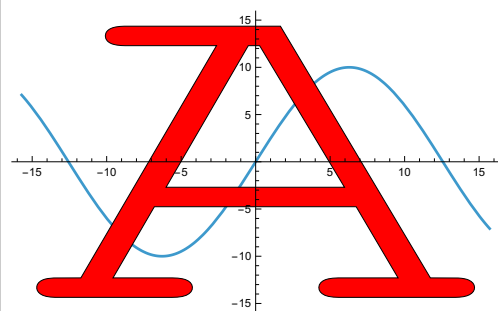
```
\begin{wolframCode}{wolfram-code-2} Ex. 28
fp1 = Plot[10 Sin[.25 x], {x, -5 Pi, 5 Pi}];

myChar = Style["A", FontFamily -> "Courier New", FontSize -> 50];
boundaryMesh = BoundaryDiscretizeGraphics[Text[myChar], _Text];
fp2 = Region[boundaryMesh, BaseStyle -> {FaceForm[Red],
```

```
EdgeForm[Black]]]

fig = Show[fp1, fp2, PlotRange->All];
Export[scriptPath <> "wolfram-code_fig.pdf", fig];
\end{wolframCode}

\includegraphics[width=.5\linewidth]{
\wolframOutputPath/wolfram-code_fig.pdf}
```



`\g_ztikz_wolfram_res_seq` ★

New: 2026-05-07

This variable stores the previous WolframScript calculation result. Ordinary users should use the subsequent `\wolframResult` command.

5.6 l3draw Library

zTikZ provides an `l3draw` library based on the `l3draw` package, mainly for simple drawing needs. For general users, it offers `\zrule` and `\zplot` commands for gradient rectangles and functions. Wrappers for `l3draw` environments and commands are also provided but are currently incomplete and not recommended for general use.

<code>\zdrawSetUnit</code>	<code>\zdrawSetUnit[<i><unit></i>]</code>
New: 2025-05-15	Sets the drawing unit, e.g., <i><unit></i> as “cm”.

<code>\zdrawSetPathWidth</code>	<code>\zdrawSetPathWidth[<i><width></i>]</code>
New: 2025-05-15	Sets the line width, e.g., <i><width></i> as “0.5pt”; the default is 0.4pt in <code>l3draw</code> .

<code>\zrule</code>	<code>\zrule[<i><key-value></i>]</code>
New: 2025-05-15	Draws a gradient rectangle with attributes set by <i><key-value></i> .

<code>ztikz/zdraw/zrule/width</code>	<code>width</code>	<code>=</code>	<code><float></code>		Initial value:	<code>1</code>
<code>ztikz/zdraw/zrule/height</code>	<code>height</code>	<code>=</code>	<code><float></code>		Initial value:	<code>1</code>
<code>ztikz/zdraw/zrule/startColor</code>	<code>startColor</code>	<code>=</code>	<code><color></code>		Initial value:	<code>red</code>
<code>ztikz/zdraw/zrule/endColor</code>	<code>endColor</code>	<code>=</code>	<code><color></code>		Initial value:	<code>blue</code>
<code>ztikz/zdraw/zrule/step</code>	<code>step</code>	<code>=</code>	<code><float></code>		Initial value:	<code>0.25</code>

<width> and *<height>* set the dimensions; *<startColor>* and *<endColor>* set the colors; *<step>* controls the gradient precision.

```
\zrule[width=10, startColor=red, step=1]
```

Ex. 29



<code>\zplot</code>	<code>\zplot[<i><key-value></i>]{<i><function></i>}</code>
New: 2025-05-15	Draws functions with horizontal or vertical gradients. <i><key-value></i> sets attributes; <i><function></i> is a valid expression.

NOTE: `\zplot` is currently unstable and may error; use with caution.

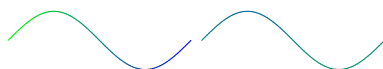
<code>ztikz/zdraw/zplot/action</code>	<code>action</code>	<code>=</code>	<code><draw stroke fill clip shade></code>		Initial value:	<code>draw</code>
<code>ztikz/zdraw/zplot/domain</code>	<code>domain</code>	<code>=</code>	<code><float, float, float></code>		Initial value:	<code>-5, 0.1, 5</code>
<code>ztikz/zdraw/zplot/range</code>	<code>range</code>	<code>=</code>	<code><float, float></code>		Initial value:	<code>-5, 5</code>
<code>ztikz/zdraw/zplot/startColor</code>	<code>startColor</code>	<code>=</code>	<code><color></code>		Initial value:	<code>black</code>
<code>ztikz/zdraw/zplot/endColor</code>	<code>endColor</code>	<code>=</code>	<code><color></code>		Initial value:	<code>white</code>
<code>ztikz/zdraw/zplot/axis</code>	<code>axis</code>	<code>=</code>	<code><x y></code>		Initial value:	<code>y</code>

<action> controls drawing behavior; *<domain>* sets the independent variable range (start, step, end); *<range>* sets the *y*-axis range, useful for *<action>*=`shade`; *<startColor>* and *<endColor>* set colors; *<axis>* sets the gradient direction (`‘x’` for horizontal, `‘y’` for vertical).

```
\def\PI{3.1415926}
```

Ex. 30

```
\zplot[
  domain={0, 0.02*\PI, 2*\PI},
  action=shade, startColor=blue,
  endColor=green, axis=x]{sin(x)}
\zplot[
  domain={0, 0.02*\PI, 2*\PI},
  action=shade, startColor=blue,
  endColor=green, axis=y]{sin(x)}
```



Zdraw `\begin{zdraw} <l3draw code> \end{zdraw}`

New: 2025-05-15 Wrapper for `\draw_begin:` and `\draw_end:`.

Zgroup `\begin{zgroup} <l3draw code> \end{zgroup}`

New: 2025-05-15 Wrapper for `\draw_path_scope_begin:` and `\draw_path_scope_end:`.

`\zmoveto` `\zmoveto{<coordinate>}`

`\zlineto` `\zlineto{<coordinate>}`

New: 2025-05-15 Moves the pen to `<coordinate>`, a valid `l3draw` coordinate expression like “1mm, 2cm+3em”.

`\zscolor` `\zscolor{<l3color>}`

`\zfcolor` `\zfcolor{<l3color>}`

New: 2025-05-15 `<l3color>` is a valid `l3draw` color expression; *z*TikZ predefines common colors as “`l3color`” variables.

`\zfevenodd` `\zfevenodd` specifies the even-odd rule for fill division, while `\zfnozero` specifies the non-zero rule.

New: 2025-05-31

`\zxvec` `\zxvec{<coordinate>}`

`\zyvec` `\zyvec{<coordinate>}`

New: 2025-05-15 Sets the unit vectors for x and y axes.

`\zpolar` `\zpolar{<radius>}{<angle>}`

`\zcoor` `\zcoor{<x-scale>}{<y-scale>}`

New: 2025-05-15 `\zpolar` gets coordinates via polar method; `\zcoor` via rectangular method relative to the base vectors.

<code>\zrect</code>	<code>\zrect{<coordinate>}{<coordinate>}</code>
<code>\zcirc</code>	<code>\zcirc{<center>}{<radius>}</code>
New: 2025-05-15	Draws rectangles (lower-left to upper-right) or circles (center and radius).
<code>\znewtext</code>	<code>\znewtext{<coffin>}</code>
<code>\zsethtext</code>	<code>\zsethtext{<coffin>}{<content>}</code>
<code>\zsetvtext</code>	<code>\zsetvtext{<coffin>}{<width>}{<content>}</code>
<code>\zscaletext</code>	<code>\zscaletext{<coffin>}{<x-scale>}{<y-scale>}</code>
<code>\zputtext</code>	<code>\zputtext{<coffin>}{<hpole>}{<vpole>}{<point>}</code>
New: 2025-05-15	Macros for creating, transforming, and placing text in <code>l3draw</code> .
<code>\zbg</code>	Wrappers for <code>\draw_path_scope_begin:</code> and <code>\draw_path_scope_end:</code> .
<code>\zeg</code>	
New: 2025-05-15	
<code>\zcapbutt</code>	Set line connection styles.
<code>\zcaproun</code>	
<code>\zcaprect</code>	
<code>\zclosepath</code>	
New: 2025-05-15	
<code>\zshift</code>	<code>\zshift{<vector>}</code>
<code>\zxscale</code>	<code>\zxscale{<x-scale>}</code>
<code>\zyscale</code>	<code>\zyscale{<y-scale>}</code>
<code>\ztrans</code>	<code>\ztrans{<a>}{}{<c>}{<d>}</code>
New: 2025-05-15	Macros for affine transformations of axes.
<code>\zusepath</code>	<code>\zusepath[<style>]</code>
New: 2025-05-15	Shows the final path; <code><style></code> defaults to “draw”, other options: “stroke, fill, clip”.

6 Appendix

6.1 gnuplot Built-in Functions

Here we supplement the built-in functions in gnuplot: Arguments to math functions in gnuplot can be integer, real, or complex unless otherwise noted. Functions that accept or return angles (e.g. $\sin(x)$) treat angle values as radians, but this may be changed to degrees using the command `set angles`. (Excerpt from: [gnuplot support functions](#))

Table 1: **gnuplot math library functions**

Function	Arguments	Returns
$\text{abs}(x)$	any	$ x $, absolute value of x ; same type
$\text{abs}(x)$	complex	length of x , $\sqrt{\text{Re}(x)^2 + \text{Im}(x)^2}$
$\text{acos}(x)$	any	$\cos^{-1} x$ (inverse cosine)
$\text{acosh}(x)$	any	$\cosh^{-1} x$ (inverse hyperbolic cosine) in radians
$\text{airy}(x)$	any	Airy function $\text{Ai}(x)$
$\text{arg}(x)$	complex	the phase of x
$\text{asin}(x)$	any	$\sin^{-1} x$ (inverse sine)
$\text{asinh}(x)$	any	$\sinh^{-1} x$ (inverse hyperbolic sine) in radians
$\text{atan}(x)$	any	$\tan^{-1} x$ (inverse tangent)
$\text{atan2}(y, x)$	int or real	$\tan^{-1}(y/x)$ (inverse tangent)
$\text{atanh}(x)$	any	$\tanh^{-1} x$ (inverse hyperbolic tangent) in radians
$\text{EllipticK}(k)$	real k in $(-1 : 1)$	$K(k)$ complete elliptic integral of the first kind
$\text{EllipticE}(k)$	real k in $[-1 : 1]$	$E(k)$ complete elliptic integral of the second kind
$\text{EllipticPi}(n, k)$	real $n, k < 1$	$\Pi(n, k)$ complete elliptic integral of the third kind
$\text{besj0}(x)$	int or real	J_0 Bessel function of x , in radians
$\text{besj1}(x)$	int or real	J_1 Bessel function of x , in radians
$\text{besy0}(x)$	int or real	Y_0 Bessel function of x , in radians
$\text{besy1}(x)$	int or real	Y_1 Bessel function of x , in radians
$\text{ceil}(x)$	any	$\lceil x \rceil$, smallest integer not less than x (real part)
$\text{cos}(x)$	radians	$\cos x$, cosine of x
$\text{cosh}(x)$	any	$\cosh x$, hyperbolic cosine of x in radians
$\text{erf}(x)$	any	$\text{erf}(\text{Re}(x))$, error function of $\text{Re}(x)$
$\text{erfc}(x)$	any	$\text{erfc}(\text{Re}(x))$, 1.0– error function of $\text{Re}(x)$
$\text{exp}(x)$	any	e^x , exponential function of x
$\text{expint}(n, x)$	any	$E_n(x)$, exponential integral function of x
$\text{floor}(x)$	any	$\lfloor x \rfloor$, largest integer not greater than x (real part)
$\text{gamma}(x)$	any	$\Gamma(\text{Re}(x))$, gamma function of $\text{Re}(x)$
$\text{ibeta}(p, q, x)$	any	$\text{ibeta}(\text{Re}(p, q, x))$, ibeta function of $\text{Re}(p, q, x)$
$\text{inverf}(x)$	any	inverse error function $\text{Re}(x)$
$\text{igamma}(a, x)$	any	$\text{igamma}(\text{Re}(a, x))$, igamma function of $\text{Re}(a, x)$
$\text{imag}(x)$	complex	$\text{Im}(x)$, imaginary part of x as a real number
$\text{invnorm}(x)$	any	inverse normal distribution function $\text{Re}(x)$
$\text{int}(x)$	real	integer part of x , truncated toward zero
$\text{lambertw}(x)$	real	Lambert W function

$\lgamma(x)$	any	$\lgamma(\text{Re}(x))$, lgamma function of $\text{Re}(x)$
$\log(x)$	any	$\ln x$, natural logarithm (base e) of x
$\log_{10}(x)$	any	$\log_{10} x$, logarithm (base 10) of x
$\text{norm}(x)$	any	$\text{norm}(x)$, normal distribution function of $\text{Re}(x)$
$\text{rand}(x)$	int	pseudo random number in the interval $(0 : 1)$
$\text{real}(x)$	any	$\text{Re}(x)$, real part of x
$\text{sgn}(x)$	any	1 if $x > 0$, -1 if $x < 0$, 0 if $x = 0$. $\Im(x)$ ignored
$\sin(x)$	any	$\sin x$, sine of x
$\sinh(x)$	any	$\sinh x$, hyperbolic sine of x in radians
$\text{sqrt}(x)$	any	$\sqrt{x}$, square root of x
$\tan(x)$	any	$\tan x$, tangent of x
$\tanh(x)$	any	$\tanh x$, hyperbolic tangent of x in radians
$\text{voigt}(x, y)$	real	convolution of Gaussian and Lorentzian
$\text{cerf}(z)$	complex	complex error function
$\text{cdawson}(z)$	complex	complex Dawson's integral
$\text{faddeeva}(z)$	complex	$w(z) = \exp(-z^2) \times \text{erfc}(-iz)$
$\text{erfi}(x)$	real	imaginary error function $\text{erfi}(x) = -i \times \text{erf}(ix)$
$\text{VP}(x, \sigma, \gamma)$	real	Voigt profile

Note: $\text{faddeeva}(z)$: rescaled complex error function

6.2 marker Styles

The marker styles available in `TikZ` are as follows:

<code>\pgfuseplotmark{-}</code>	
<code>\pgfuseplotmark{ }</code>	
<code>\pgfuseplotmark{o}</code>	
<code>\pgfuseplotmark{asterisk}</code>	
<code>\pgfuseplotmark{star}</code>	
<code>\pgfuseplotmark{10-pointed star}</code>	
<code>\pgfuseplotmark{oplus}</code>	
<code>\pgfuseplotmark{oplus*}</code>	
<code>\pgfuseplotmark{otimes}</code>	
<code>\pgfuseplotmark{otimes*}</code>	
<code>\pgfuseplotmark{square}</code>	
<code>\pgfuseplotmark{square*}</code>	
<code>\pgfuseplotmark{triangle}</code>	
<code>\pgfuseplotmark{triangle*}</code>	
<code>\pgfuseplotmark{diamond}</code>	
<code>\pgfuseplotmark{diamond*}</code>	
<code>\pgfuseplotmark{halfdiamond*}</code>	
<code>\pgfuseplotmark{halfsquare*}</code>	
<code>\pgfuseplotmark{halfsquare right*}</code>	
<code>\pgfuseplotmark{halfsquare left*}</code>	
<code>\pgfuseplotmark{pentagon}</code>	
<code>\pgfuseplotmark{pentagon*}</code>	
<code>\pgfuseplotmark{Mercedes star}</code>	
<code>\pgfuseplotmark{Mercedes star flipped}</code>	
<code>\pgfuseplotmark{halfcircle}</code>	
<code>\pgfuseplotmark{halfcircle*}</code>	
<code>\pgfuseplotmark{heart}</code>	
<code>\pgfuseplotmark{text}</code>	

Figure 3: `TikZ` Marker Styles

6.3 Test Data/Code

```
# Curve 0 of 1, 10 points
# Curve title: "f(x)"
# x y type
-3.14159 -0.00000 i
-2.44346 -0.64279 i
-1.74533 -0.98481 i
-1.04720 -0.86603 i
-0.34907 -0.34202 i
0.34907 0.34202 i
1.04720 0.86603 i
1.74533 0.98481 i
2.44346 0.64279 i
3.14159 0.00000 i
```

sine.data

```
\begin{pycode}{pycode_table.txt}
import numpy as np

# write file
with open ('pycode_table.txt', 'w') as file:
    file.write("\begin{tabular}{p{3cm}ccc}\n")
    file.write("\hline\n")
    file.write("number/function & $\sin$ & $\cos$ & $\tan$\\n")
\n")
    file.write("\hline\n")
    for i in range(1, 16):
        file.write(
            f"${i}$ & ${np.around(np.sin(i), decimals=4)}$ &
${np.around(np.cos(i), decimals=4)}$ & ${np.around(np.tan(i),
decimals=4)}$\\n"
        )

    file.write("\hline\n")
    file.write("\end{tabular}\n")
\end{pycode}
```

table.py.txt

7 TODO

Development of $\text{\texttt{\textit{tikz}}}$ has come to a pause for now. Here are some functions that may be added in the future (☐ – undone; ☒ – done; ☒ – will not fix):

- ☒ (luadraw package already exists) Implement an interface similar to `tikz-3dplot`, rewriting it with $\text{\texttt{\textit{L}A}T\text{\texttt{\textit{E}}}X3$.
- ☒ (Refer to processing method of the “`wolframAny`” environment) Add calling interfaces for Matlab scripts, or directly use its open-source alternative `GNU Octave`?
- ☒ (Refer to implementation of “`mermaid`” environment in cache library) Implement the `wolframAny` environment, providing functionality similar to `pycode`.
- ☒ 2025-05-29-Done: Rewrite the function `\ztikz_hash_if_change:nn` for the cache mechanism, as it is currently not flexible enough (maybe directly use the `robust-externalize` package).
- ☒ 2025-05-29-Done: For the cache library, redundant Hash values need to be cleared: for example, if an environment/command generates an original Hash value of “A”, and after parameters change, the Hash becomes “B”, then the original “A” needs to be cleared.
- ☒ 2025-05-29-Done: Implement the `\ztikzForceToSkip` command: how to handle the `\xsim_file_write_start:nn` environment? How to handle multiple `\ztikzForceToRun`? Maybe using `\str_mdfive_hash:n` would be helpful?
- ☐ Integrate more features related to `wstp`: Take over $\text{\texttt{\textit{T}}}E\text{\texttt{\textit{X}}}$ low-level pagination and box calculations (global optimum?), index and custom sort generation, complex regular expressions, and metaprogramming.

8.1 ztikz.sty

```

1 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% 1
2 % ztikz.sty % 2
3 % Copyright 2024, 2025, 2026 Zongping Ding. % 3
4 % % 4
5 % This work may be distributed and/or modified under the conditions of the % 5
6 % LaTeX Project Public License, either version 1.3 of this license or any % 6
7 % later version. % 7
8 % The latest version of this license is in % 8
9 % http://www.latex-project.org/lppl.txt % 9
10 % and version 1.3 or later is part of all distributions of LaTeX % 10
11 % version 2005/12/01 or later. % 11
12 % % 12
13 % This work has the LPPL maintenance status `maintained'. % 13
14 % % 14
15 % The Current Maintainer of this work is Zongping Ding. % 15
16 % % 16
17 % This work consists of the files ztikz.sty, % 17
18 % the libraries: ztikz.library.gnuplot.tex, % 18
19 % ztikz.library.cache.tex, % 19
20 % ztikz.library.python.tex, % 20
21 % ztikz.library.wolfram.tex, % 21
22 % ztikz.library.l3draw.tex, % 22
23 % and the script files: ztikz.library.gnuscrypt.tex, % 23
24 % ztikz.library.pyscript.tex. % 24
25 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% 25
26 \NeedsTeXFormat{LaTeX2e} 26
27 \ProvidesExplPackage{ztikz}{2026/05/08} 27
28 {1.1.0}{A~pre-release~diagram~and~CAS~package} 28
29 29
30 30
31 % ----- 31
32 % basic tools 32
33 % ----- 33
34 \RequirePackage{ztool} 34
35 \RequirePackage{graphicx} 35
36 \__ztool_load_library:n {shell-escape, file-io} 36
37 \cs_new_protected:Npn \ztikz_hook_preamble_last:n #1 37
38 { \AddToHook{env/document/before}{#1} } 38
39 \cs_new_protected:Npn \ztikz_label_hook_preamble_last:nn #1#2 39
40 { \AddToHook{env/document/before}[#1]{#2} } 40
41 \cs_new_protected:Npn \ztikz_hook_doc_begin:n #1 41
42 { \AddToHook{begindocument}{#1} } 42
43 \cs_new_protected:Npn \ztikz_hook_doc_end:n #1 43
44 { \AddToHook{enddocument}{#1} } 44
45 45

```

46		46
47		47
48	% -----	48
49	%	49
50	% -----	50
51	\clist_new:N \g__ztikz_library_loaded_clist	51
52	\clist_gclear:N \g__ztikz_library_loaded_clist	52
53	\cs_new_nopar:Npn __ztikz_load_library:n #1	53
54	{	54
55	\clist_map_inline:nn {#1} {	55
56	\clist_if_in:NnTF \g__ztikz_library_loaded_clist {##1} {	56
57	\msg_set:nnn {ztikz} {library-loaded}	57
58	{	58
59	ztikz~library~"##1"~already~loaded,ignored~loading.	59
60	\msg_line_context:	60
61	}	61
62	\msg_warning:nnn {ztikz} {library-loaded} {##1}	62
63	}{	63
64	\file_if_exist:nTF {library/ztikz.library.##1.tex}{	64
65	\clist_gput_right:Nn \g__ztikz_library_loaded_clist {##1}	65
66	\makeatletter\file_input:n {library/ztikz.library.##1.tex}	66
67	}{	67
68	\msg_set:nnn {ztikz} {library-not-found} {ztikz~library~`##1'~not~found.}	68
69	\msg_error:nnn {ztikz} {library-not-found} {##1}	69
70	}	70
71	}	71
72	}	72
73		73
74	\NewDocumentCommand\ztikzloadlib{m}	74
75	{	75
76	__ztikz_load_library:n {#1}	76
77	\ExplSyntaxOff	77
78	}	78
79		79
80		80
81		81
82	% -----	82
83	%	83
84	% -----	84
85	\RequirePackage{xsimverb}	85
86	\cs_generate_variant:Nn \xsim_file_write_start:nn {ne}	86
87	\ztool_shell_mkdir:n {ztikz_output/}	87
88	\tl_const:Nn \g__ztikz_scripts_path_tl {ztikz_output/scripts}	88
89	\NewDocumentCommand\ztikzMkdir{m}{ \ztool_shell_mkdir:n {#1} }	89
90		90
91		91
92		92
93	% -----	93

```
94 %                                ztikz's options                                94
95 % -----                                95
96 \cs_new_protected:Npn \ztikz_keys_define:n #1                                96
97   { \keys_define:nn { ztikz }{ #1 } }                                97
98 \cs_new_protected:Npn \ztikz_keys_define:nn #1#2                                98
99   { \keys_define:nn { ztikz / #1 }{ #2 } }                                99
100 \cs_new_protected:Npn \ztikz_keys_set:nn #1#2                                100
101   { \keys_set:nn { ztikz / #1 }{ #2 } }                                101
102 \ztikz_keys_define:n                                102
103   {                                103
104     quiet                .bool_gset:N = \g_ztikz_quiet_bool,                                104
105     quiet                .initial:n   = { false },                                105
106     quiet                .default:n   = { true },                                106
107     library              .multichoice:,                                107
108     library / basic      .code:n       = { \__ztikz_load_library:n {basic} },                                108
109     library / gnuplot     .code:n       = { \__ztikz_load_library:n {gnuplot} },                                109
110     library / cache      .code:n       = { \__ztikz_load_library:n {cache} },                                110
111     library / python      .code:n       = { \__ztikz_load_library:n {python} },                                111
112     library / wolfram     .code:n       = { \__ztikz_load_library:n {wolfram} },                                112
113     library / l3draw      .code:n       = { \__ztikz_load_library:n {l3draw} },                                113
114     library / unknown     .code:n       = {                                114
115       \msg_set:nnn {ztikz} {library-not-found}                                115
116       {ztikz~library~`#1'~not~found,~valid~libraries~are~'basic',                                116
117       ~'gnuplot',~'cache',~'python',~'wolfram'~and~'l3draw'.}                                117
118       \msg_error:nn {ztikz} {library-not-found}                                118
119     },                                119
120     wolfram              .meta:nn      = { ztikz/wolfram }{ #1 },                                120
121   }                                121
122                                122
123 % wolfram meta-key                                123
124 \tl_new:N \g__ztikz_wolfram_engine_tl                                124
125 \bool_new:N \g__ztikz_wolfram_cloud_bool                                125
126 \ztikz_keys_define:nn { wolfram }                                126
127   {                                127
128     engine .choice:,                                128
129     engine / wolfram .code:n =                                129
130       {                                130
131         \tl_gset:Nn \g__ztikz_wolfram_engine_tl { wolframscript }                                131
132       },                                132
133     engine / mathics .code:n =                                133
134       {                                134
135         \tl_gset:Nn \g__ztikz_wolfram_engine_tl { mathics }                                135
136       },                                136
137     engine / wstp .code:n =                                137
138       {                                138
139         \tl_gset:Nn \g__ztikz_wolfram_engine_tl { wstp }                                139
140       },                                140
141     engine / unknown .code:n =                                141
```

142	{	142
143	\msg_set:nnn {ztikz} {wolfram-engine-not-found}	143
144	{ztikz's~'wolfram'~engine~option~`#1'~invalid,	144
145	~valid~engine~is~'wolframscript'~or~'mathics'.}	145
146	\msg_error:nn {ztikz} {wolfram-engine-not-found}	146
147	},	147
148	engine .initial:n = { wolfram },	148
149	cloud .choices:nn = { true, false }	149
150	{	150
151	\use:c { bool_gset_ \l_keys_choice_tl :N }	151
152	\g__ztikz_wolfram_cloud_bool	152
153	},	153
154	cloud .initial:n = { false },	154
155	cloud .default:n = { true },	155
156	}	156
157	\ProcessKeyOptions [ztikz]	157
158		158
159	% only 'wolfram' support 'cloud'	159
160	\bool_if:NT \g__ztikz_wolfram_cloud_bool	160
161	{	161
162	\tl_if_eq:VnT \g__ztikz_wolfram_engine_tl { mathics }	162
163	{	163
164	\msg_set:nnn {ztikz}{mathics-cloud}	164
165	{mathics~'cloud'~is~not~supported.}	165
166	\msg_error:nn {ztikz}{mathics-cloud}	166
167	}	167
168	}	168
169		169
170	% whether to show these debug info	170
171	\bool_if:NTF \g_ztikz_quiet_bool	171
172	{	172
173	\cs_new_protected:Npn \ztikz_term_info:n #1	173
174	{	174
175	}	175
176	\cs_new_protected:Npn \ztikz_term_info:e #1	176
177	{	177
178	}	178
179	}{	179
180	\cs_new_protected:Npn \ztikz_term_info:n #1	180
181	{	181
182	\iow_now:Nn \c_term_iow {#1}	182
183	}	183
184	\cs_new_protected:Npn \ztikz_term_info:e #1	184
185	{	185
186	\iow_now:Ne \c_term_iow {#1}	186
187	}	187
188	}	188
189		189

```

190
191
192 % -----
193 %
194 % -----
195 \cs_generate_variant:Nn \ior_open:Nn { Ne }
196 \cs_generate_variant:Nn \iow_open:Nn { Ne }
197 \cs_generate_variant:Nn \tl_map_function:nN { eN }
198 \ztikz_hook_preamble_last:n
199 {
200     \clist_if_in:NnF \g__ztikz_library_loaded_clist {cache}
201     {
202         \tl_new:N \l__ztikz_current_hash_tl
203         \prg_new_conditional:Npnn \ztikz_if_run_again:nnn #1#2#3 {p, T, F, TF}
204         {
205             \bool_gset_true:N \g__hash_change_bool
206             \prg_return_true:
207         }
208         \prg_generate_conditional_variant:Nnn \ztikz_if_run_again:nnn
209         { nen } { p, T, F, TF }
210     }
211 }
212 % sed script
213 \cs_new:Npn \__ztikz_addto_script:nn #1#2
214 {% #1: file; #2: line; #3:new
215     \ztool_append_to_file:nn
216     {#1}{#2}
217 }
218 \cs_generate_variant:Nn \__ztikz_addto_script:nn { ne, en, ee }
219 \ztikz_keys_define:nn { sed-script }
220 {
221     file .tl_set:N = \l__ztikz_sed_script_file_tl,
222     line .int_set:N = \l__ztikz_sed_script_line_int,
223     ori .tl_set:N = \l__ztikz_sed_script_ori_tl,
224     new .tl_set:N = \l__ztikz_sed_script_new_tl,
225 }
226 \cs_new:Npn \__ztikz_sed_script:nnn #1#2#3
227 {% #1: file; #2: line; #3:new
228     \ztool_replace_file_line:nnn
229     {\g__ztikz_scripts_path_tl/#1}
230     {#2}{#3}
231 }
232 \cs_generate_variant:Nn \__ztikz_sed_script:nnn {eee}
233 \cs_generate_variant:Nn \__ztikz_sed_script:nnn {nne}
234 % ztik logo
235 \NewDocumentCommand\TikZ{}{Ti\textcolor{orange}{\textit{k}}Z}
236 \NewDocumentCommand\zTikZ{}
237 {

```

238	<code>\ztool_scale_to_wd_and_ht:nnn {.9ex}{1.3ex}{</code>	238
239	<code>\ztool_rotate:nn {89}{\(\aleph\)}</code>	239
240	<code>}\kern-0.3423ex\hbox{\TikZ</code>	240
241	<code>}</code>	241
242	<code>\let\ztikz\zTikZ</code>	242
243	<code>\protected\def\HoLogo@zTikZ#1{\zTikZ</code>	243
244	<code>\protected\def\HoLogo@ztikz#1{\zTikZ</code>	244

8.2 Library

8.2.1 basic

```
1 \ProvidesExplFile{ztikz.library.basic.tex}{2025/09/22}
2 {1.0.0}{basic~library~for~ztikz}
3
4
5
6 % -----
7 %                               basic packages
8 % -----
9 \RequirePackage{tikz}
10 \RequirePackage{etoolbox}
11 \patchcmd{\pgfutil@InputIfFileExists}{\input #1}{%
12   \@pushfilename
13   \xdef\@currname{#1}
14   \input #1
15   \@popfilename
16 }{}{}
17 \usetikzlibrary{arrows.meta}
18 \usetikzlibrary{intersections}
19 \usetikzlibrary{patterns}
20 \usetikzlibrary{plotmarks}
21 \usetikzlibrary{positioning}
22 \usetikzlibrary{shapes.geometric}
23 \usetikzlibrary{decorations.markings}
24 \usetikzlibrary{fadings}
25
26
27
28 % ==> coordinate basic components
29 \ztikz_keys_define:nn { point }
30 {
31   type      .str_set:N = \l__point_type_str,
32   type      .initial:n = { * },
33   radius    .dim_set:N = \l__point_radius_dim,
34   radius    .initial:n = { 1pt },
35   color     .tl_set:N = \l__point_color_tl,
36   color     .initial:n = { black },
37   opacity   .tl_set:N = \l__point_opacity_tl,
38   opacity   .initial:n = { 1 },
39   rotate    .fp_set:N = \l__point_rotate_angle,
40   rotate    .initial:n = { 0 },
41 }
42 \NewDocumentCommand\ShowPoint{ 0{}m0{}0{} }
43 {
44   \group_begin:
45   \exp_args:Nne \ztikz_keys_set:nn { point } { #1 }
```

```

46 \seq_set_split:Nnn \l__point_list_seq { ; }{#2}
47 \seq_set_split:Nnn \l__point_label_seq { ; }{#3}
48 \int_step_inline:nnnn {1}{1}
49 { \seq_count:N \l__point_list_seq }
50 {
51 \draw plot [
52 only~ marks,
53 mark = \str_use:N \l__point_type_str,
54 mark~ size = \dim_use:N \l__point_radius_dim,
55 mark~ options = {
56 rotate = \fp_use:N \l__point_rotate_angle,
57 opacity = \tl_use:N \l__point_opacity_tl,
58 color = \tl_use:N \l__point_color_tl,
59 ball~ color = \tl_use:N \l__point_color_tl,
60 }
61 ] coordinates{\seq_item:Nn \l__point_list_seq{##1}}
62 node[#4] {\seq_item:Nn \l__point_label_seq{##1}};
63 }
64 \group_end:
65 }
66 \NewDocumentCommand\ShowGrid{ 0{color=gray, very~ thin, step=1}m }
67 {
68 \seq_set_split:Nnn \l__grid_param_ii_seq { ; }{#2}
69 \draw[#1] \seq_item:Nn \l__grid_param_ii_seq{1}
70 grid \seq_item:Nn \l__grid_param_ii_seq{2};
71 }
72 % intersection
73 \NewDocumentCommand\ShowIntersection{ omm }
74 {
75 \seq_set_split:Nnn \l__intersection_num_seq { ; }{#2}
76 \path[name~ intersections={
77 of=\seq_item:Nn \l__intersection_num_seq{1}~
78 and~ \seq_item:Nn \l__intersection_num_seq{2}
79 }];
80 \int_step_inline:nnnn {1}{1}{#3}{
81 \ShowPoint[#1]{(intersection-##1)}
82 }
83 }
84 % polygon plot
85 \ztikz_keys_define:nn { polygon }
86 {
87 radius .fp_set:N = \l__polygon_radius_fp,
88 radius .initial:n = { 1 },
89 edgeColor .tl_set:N = \l__polygon_edge_color_tl,
90 edgeColor .initial:n = { black },
91 fillColor .tl_set:N = \l__polygon_fill_color_tl,
92 fillColor .initial:n = { },
93 fillOpacity .fp_set:N = \l__polygon_fill_opacity_fp,

```

```

94 fillOpacity .initial:n = { 0 },
95 rotate .fp_set:N = \l__polygon_rotate_angle,
96 rotate .initial:n = { 0 },
97 shift .tl_set:N = \l__polygon_shift_tl,
98 shift .initial:n = { (0,0) },
99 marker .tl_set:N = \l__polygon_marker_option_tl,
100 marker .initial:n = { },
101 }
102 \tl_new:N \l__ztikz_basic_poly_path_tl
103 \NewDocumentCommand\Polygon{ 0{}m }
104 {
105 \group_begin:
106 \ztikz_keys_set:nn { polygon } { #1 }
107 % strip '(' and ')'
108 \tl_replace_once:Nnn \l__polygon_shift_tl{({}}
109 \tl_replace_once:Nnn \l__polygon_shift_tl{)}}{}
110 \coordinate (mv) at (\tl_use:N \l__polygon_shift_tl);
111 % create polygon
112 \begin{scope}[shift=(mv), rotate=\fp_use:N \l__polygon_rotate_angle]
113 % arg require: #2 $\ge$ 3
114 \int_step_inline:nnn {1}{#2}{
115 % draw edges
116 \fp_set:Nn \l_angle_fp {360/#2*##1*\c_one_degree_fp}
117 \fp_set:Nn \l_angle_next_fp {360/#2*(##1+1)*\c_one_degree_fp}
118 \draw [\tl_use:N \l__polygon_edge_color_tl]
119 ( \fp_eval:n {\l__polygon_radius_fp*cos(\l_angle_fp)},
120 \fp_eval:n {\l__polygon_radius_fp*sin(\l_angle_fp)}
121 ) -- (
122 \fp_eval:n {\l__polygon_radius_fp*cos(\l_angle_next_fp)},
123 \fp_eval:n {\l__polygon_radius_fp*sin(\l_angle_next_fp)}
124 );
125 % fill polygon path
126 \int_compare:nNnTF {##1}<{#2}
127 {
128 \tl_put_right:Nn \l__ztikz_basic_poly_path_tl
129 { (p##1)-- }
130 }{
131 \tl_put_right:Nn \l__ztikz_basic_poly_path_tl
132 { (p##1)--cycle }
133 }
134 % mark coordinates
135 \coordinate (p##1) at (
136 \fp_eval:n {\l__polygon_radius_fp*cos(\l_angle_fp)},
137 \fp_eval:n {\l__polygon_radius_fp*sin(\l_angle_fp)}
138 );
139 }
140 % fill polygon (none-color -> opacity=1; or opacity=.75)
141 \tl_if_empty:NTF \l__polygon_fill_color_tl {

```

```

142 \fp_set:Nn \l__polygon_fill_opacity_fp {0}
143 }{
144 \fp_set:Nn \l__polygon_fill_opacity_fp {.75}
145 }
146 \fill [
147 \tl_use:N \l__polygon_fill_color_tl,
148 fill~opacity=\fp_use:N \l__polygon_fill_opacity_fp
149 ] \l__ztikz_basic_poly_path_tl;
150 % show markers
151 \int_step_inline:nnn {1}{#2}
152 {
153 \ShowPoint[\l__polygon_marker_option_tl]
154 { (p##1) }
155 }
156 \end{scope}
157 \group_end:
158 }
159
160
161 % ==> axis
162 \ztikz_keys_define:nn { axis }
163 {
164 % basic tick args
165 tickStart .fp_set:N = \l__start_fp,
166 tickStart .initial:n = { -5 },
167 tickEnd .fp_set:N = \l__end_fp,
168 tickEnd .initial:n = { 5 },
169 axisRotate .fp_set:N = \l__axis_rotate_angle,
170 axisRotate .initial:n = { 0 },
171 % tick dimension spec
172 mainStep .fp_set:N = \l__main_step_fp,
173 mainStep .initial:n = { 1.0 },
174 subStep .fp_set:N = \l__sub_step_fp,
175 subStep .initial:n = { 0.1 },
176 mainTickLabel .tl_set:N = \l__main_tick_label_tl,
177 mainTickLabel .initial:n = { \fp_use:N {\CurrentFp} },
178 tickLabelShift .dim_set:N = \l__tick_label_shift_dim,
179 tickLabelShift .initial:n = { 0pt },
180 mainTickLength .dim_set:N = \l__main_tick_length_dim,
181 mainTickLength .initial:n = { 4pt },
182 subTickLength .dim_set:N = \l__sub_tick_length_dim,
183 subTickLength .initial:n = { 2pt },
184 mainTickLabelPosition .tl_set:N = \l__main_tick_label_position_tl,
185 mainTickLabelPosition .initial:n = { below },
186 % color spec
187 axisColor .tl_set:N = \l__axis_color_tl,
188 axisColor .initial:n = { black },
189 mainTickColor .tl_set:N = \l__main_tick_color_tl,

```

```

190     mainTickColor      .initial:n    = { black },
191     subTickColor       .tl_set:N     = \l__sub_tick_color_tl,
192     subTickColor       .initial:n    = { black },
193     mainTickLabelColor .tl_set:N     = \l__main_tick_label_color_tl,
194     mainTickLabelColor .initial:n    = { black },
195     % tick cross type spec
196     tickStyle          .choice:,
197     tickStyle/cross    .code:n       = \tl_set:Nn \l__tick_spec_tl { cross },
198     tickStyle/above    .code:n       = \tl_set:Nn \l__tick_spec_tl { above },
199     tickStyle/below    .code:n       = \tl_set:Nn \l__tick_spec_tl { below },
200 }
201 % ticks style
202 \tl_new:N \l__tick_type_tl    % `main' or `sub'
203 \tl_new:N \l__tick_spec_tl    % `cross', `above' or `below'
204 \tl_new:N \l__tick_color_tl
205 \dim_new:N \l__tick_length_dim
206 \tl_new:N \l__node_text_tl
207 % draw ticks (main or sub)
208 \cs_new_protected:Npn \ztikz_draw_axis_ticks_cs:n #1
209 {
210     \str_case:NnT \l__tick_type_tl {
211         {main}{
212             \dim_set_eq:NN \l__tick_length_dim
213                 \l__main_tick_length_dim
214             \tl_set:NV \l__tick_color_tl
215                 \l__main_tick_color_tl
216             \tl_set:Nn \l__node_text_tl
217                 { \tl_use:N \l__main_tick_label_tl }
218         }
219         {sub}{
220             \dim_set_eq:NN \l__tick_length_dim \l__sub_tick_length_dim
221             \tl_set:NV \l__tick_color_tl \l__sub_tick_color_tl
222             \tl_set:Nn \l__node_text_tl {}
223         }
224     }{}
225     \str_case:VnT \l__tick_spec_tl {
226         {cross}{
227             \draw[\tl_use:N \l__tick_color_tl]
228                 (#1, 0)++(0, \dim_eval:n {\l__tick_length_dim/2})
229                 -- ++(0, \dim_eval:n {-\l__tick_length_dim})
230                 node[\tl_use:N \l__main_tick_label_position_tl]
231                 {
232                     \textcolor{\tl_use:N \l__main_tick_label_color_tl}
233                         {\tl_use:N \l__node_text_tl}
234                 };
235         }
236         {above}{
237             \draw[\tl_use:N \l__tick_color_tl] (#1, 0)

```

```

238      -- ++(0, \dim_eval:n {\l__tick_length_dim/2})
239      node[\tl_use:N \l__main_tick_label_position_tl]
240      {
241          \textcolor{\tl_use:N \l__main_tick_label_color_tl}
242          {\tl_use:N \l__node_text_tl}
243      };
244  }
245  {below}{
246      \draw[\tl_use:N \l__tick_color_tl] (#1, 0)
247      -- ++(0, \dim_eval:n {-\l__tick_length_dim/2})
248      node[\tl_use:N \l__main_tick_label_position_tl
249      =\dim_use:N \l__tick_label_shift_dim]
250      {
251          \textcolor{\tl_use:N \l__main_tick_label_color_tl}
252          {\tl_use:N \l__node_text_tl}
253      };
254  }
255  }{}
256  }
257  % draw axis
258  \fp_new:N \CurrentFp
259  \int_new:N \l__substep_num_int
260  \NewDocumentCommand\ShowAxis{O{}m}
261  {
262      \group_begin:
263      \ztikz_keys_set:nn { axis } { #1 }
264      \seq_set_split:Nnn \l__points_seq { ; }{#2}
265      \begin{scope}[rotate=\fp_use:N \l__axis_rotate_angle]
266      \draw[->, \tl_use:N \l__axis_color_tl]
267          \seq_item:Nn \l__points_seq{1}
268          -- \seq_item:Nn \l__points_seq{2};
269      % draw ticks
270      \fp_step_inline:nnnn
271          {\fp_eval:n {\l__start_fp}}
272          {\fp_use:N \l__main_step_fp}
273          {\fp_use:N \l__end_fp}
274      {
275          % main ticks
276          \tl_set:Nn \l__tick_type_tl {main}
277          \fp_gset:Nn \CurrentFp {##1}
278          \ztikz_draw_axis_ticks_cs:n {##1}
279          % sub ticks
280          \tl_set:Nn \l__tick_type_tl {sub}
281          \int_set:Nn \l__substep_num_int
282              {\fp_eval:n {floor(\l__main_step_fp/\l__sub_step_fp)}}
283          \fp_compare:nNnTF {##1}<{\fp_eval:n {floor(\l__end_fp)}}{
284              \fp_step_function:nnnN
285              {\fp_eval:n {##1+\l__sub_step_fp}}

```

```

286         {\fp_use:N \l__sub_step_fp}
287         {\fp_eval:n {\#1+\l__substep_num_int*\l__sub_step_fp}}
288         \ztikz_draw_axis_ticks_cs:n
289     }{}
290 }
291 \end{scope}
292 \group_end:
293 }
294 \NewDocumentCommand{\xAxis}{0{-2}0{8}}
295 {
296     \ShowAxis[
297         tickStart=\fp_eval:n {\#1+1},
298         tickEnd=\fp_eval:n {\#2-0.75},
299         mainTickLabelPosition=below,
300         mainStep=1,          subStep=.25,
301         axisRotate=0,        axisColor=black,
302         mainTickColor=black, subTickColor=black,
303         mainTickLength=10pt, subTickLength=5pt,
304         tickLabelShift=0pt,  tickStyle=below,
305     ]{(\#1, 0); (\#2, 0)}
306 }
307 \NewDocumentCommand{\yAxis}{0{-2}0{8}}
308 {
309     \ShowAxis[
310         tickStart=\fp_eval:n {\#1+1},
311         tickEnd=\fp_eval:n {\#2-0.75},
312         mainStep=1,          subStep=.25,
313         axisRotate=90,        axisColor=black,
314         mainTickColor=black, subTickColor=black,
315         mainTickLength=10pt, subTickLength=5pt,
316         tickLabelShift=0pt,  tickStyle=above,
317         mainTickLabelPosition=left
318     ]{(\#1, 0); (\#2, 0)}
319 }
320
321
322 % ==> statistic plot function
323 \cs_new_protected:Npn \ztikz_statistic_plot_cs:nnnn #1#2#3#4
324 {% #1:starts option; #2:draw-keyval; #3:point-keyval; #4:filename
325     \tl_if_empty:nTF {#3}{\draw[#2] plot[#1] file {#4};}
326     {
327         \group_begin:
328         \keys_set:nn { ztikz / point } { #3 }
329         \draw[#2] plot [
330             % stairs options
331             #1,
332             % marker options
333             mark = \str_use:N \l__point_type_str,

```

```

334 mark~ size = \dim_use:N \l__point_radius_dim,
335 mark~ options = {
336     rotate = \fp_use:N \l__point_rotate_angle,
337     opacity = \tl_use:N \l__point_opacity_tl,
338     color = \tl_use:N \l__point_color_tl,
339     ball~ color = \tl_use:N \l__point_color_tl,
340 }
341 ] file {#4};
342 \group_end:
343 }
344 }
345 \cs_generate_variant:Nn \ztikz_statistic_plot_cs:nnnn {ennn}
346
347 % stairs plot
348 \seq_new:N \l__statistic_option_tl
349 \NewDocumentCommand\StairsPlot
350 { 0{plot-left;jump-left}0{color=black}0{}m }
351 {
352     \seq_set_split:Nnn \l__statistic_option_tl { ; }{#1}
353     \str_case:enF {\seq_item:Nn \l__statistic_option_tl{1}}{
354         {plot-left}{\tl_set:Nn \l__tmpa_tl {const~plot~mark~left}}
355         {plot-right}{\tl_set:Nn \l__tmpa_tl {const~plot~mark~right}}
356         {plot-mid}{\tl_set:Nn \l__tmpa_tl {const~plot~mark~mid}}
357         {}{\tl_set:Nn \l__tmpa_tl {}}
358     }{
359         \msg_new:nnn {ztikz}{ztikz-stairs-plot}
360         {current~stairs~plot~type~is:~'#1'~,~ invalide}
361         \msg_error:nn {ztikz}{ztikz-stairs-plot}
362     }
363     \str_case:enF {\seq_item:Nn \l__statistic_option_tl{2}}{
364         {jump-left}{\tl_set:Nn \l__tmpb_tl {jump~mark~left}}
365         {jump-right}{\tl_set:Nn \l__tmpb_tl {jump~mark~right}}
366         {jump-mid}{\tl_set:Nn \l__tmpb_tl {jump~mark~mid}}
367         {}{\tl_set:Nn \l__tmpb_tl {}}
368     }{
369         \msg_new:nnn {ztikz}{ztikz-stairs-plot}
370         {current~stairs~jump~type~is:~'#1'~,~ invalide}
371         \msg_error:nn {ztikz}{ztikz-stairs-plot}
372     }
373     \ztikz_statistic_plot_cs:ennn {\l__tmpa_tl,\l__tmpb_tl}
374     {#2}{#3}{#4}
375 }
376 % stem plot
377 \NewDocumentCommand\StemPlot{ 0{x}0{color=black}0{}m }
378 {
379     \str_case:enF {#1}{
380         {x}{\tl_set:Nn \l__tmpa_tl {ycomb}}
381         {y}{\tl_set:Nn \l__tmpa_tl {xcomb}}

```

382	<code>{o}{\tl_set:Nn \l__tmpa_tl {polar~ comb}}</code>	382
383	<code>{\tl_set:Nn \l__tmpa_tl {ycomb}}</code>	383
384	<code>{}</code>	384
385	<code>\msg_new:nnn {ztikz}{ztikz-stem-plot}</code>	385
386	<code>{current~stem~plot~type~is::~'#1'~,~ invalide}</code>	386
387	<code>\msg_error:nn {ztikz}{ztikz-stem-plot}</code>	387
388	<code>}</code>	388
389	<code>\ztikz_statistic_plot_cs:ennn {\l__tmpa_tl}{#2}{#3}{#4}</code>	389
390	<code>}</code>	390
391	<code>% bar plot</code>	391
392	<code>\NewDocumentCommand\BarPlot{ 0{ybar}0{color=black}0{m }}</code>	392
393	<code>{</code>	393
394	<code>\str_case:enF {#1}{</code>	394
395	<code>{x}{\tl_set:Nn \l__tmpa_tl {ybar}}</code>	395
396	<code>{y}{\tl_set:Nn \l__tmpa_tl {xbar}}</code>	396
397	<code>{xc}{\tl_set:Nn \l__tmpa_tl {ybar~ interval}}</code>	397
398	<code>{yc}{\tl_set:Nn \l__tmpa_tl {xbar~ interval}}</code>	398
399	<code>{\tl_set:Nn \l__tmpa_tl {ybar}}</code>	399
400	<code>}</code>	400
401	<code>\msg_new:nnn {ztikz}{ztikz-bar-plot}</code>	401
402	<code>{current~bar~plot~type~is::~'#1'~,~ invalide}</code>	402
403	<code>\msg_error:nn {ztikz}{ztikz-bar-plot}</code>	403
404	<code>}</code>	404
405	<code>\ztikz_statistic_plot_cs:ennn {\l__tmpa_tl}{#2}{#3}{#4}</code>	405
406	<code>}</code>	406

8.2.2 gnuplot

```
1 \ProvidesExplFile{ztikz.library.gnuplot.tex}{2025/12/02}
2 {1.0.0}{gnuplot~library~for~ztikz}
3
4
5
6 % ==> init variables
7 \ztool_shell_mkdir:n {ztikz_output/gnuplot_data/}
8 \__ztikz_load_library:n {gnuscript}
9 \int_new:N \g__tikz_env_index_int
10 \int_new:N \g__gnu_data_index_int
11 \int_new:N \g__gnu_plotz_index_int
12 \int_gadd:Nn \g__gnu_plotz_index_int {1}
13 \tl_const:Nn \g__ztikz_gnu_path_tl {ztikz_output/gnuplot_data}
14 \pretocmd{\tikzpicture}{
15   \int_gincr:N \g__tikz_env_index_int
16   \int_gset:Nn \g__gnu_data_index_int {0}
17 }{}{}
18
19 % plot args
20 \tl_new:N \l__pairs_x_domain_tl
21 \tl_new:N \l__pairs_y_domain_tl
22 \ztikz_keys_define:nn { 2dplot }
23 {
24   domain .tl_set:N = \l__ztikz_plot_domain_tl,
25   domain .initial:n = { -5:5; },
26   style .tl_set:N = \l__ztikz_plot_style_tl,
27   marker .tl_set:N = \l__ztikz_plot_marker_tl,
28 }
29 \ztikz_keys_define:nn { 3dplot }
30 {
31   domain .tl_set:N = \l__ztikz_plotz_domain_tl,
32   domain .initial:n = {-5:5; -5:5},
33   palette .tl_set:N = \l__ztikz_plotz_palette_tl,
34   palette .initial:n = {rgbformulae~ 22,13,-31},
35   pm3d .bool_set:N = \l__ztikz_plotz_pm_bool,
36   pm3d .initial:n = {true},
37   pm3d .default:n = {true},
38   width .dim_set:N = \l__ztikz_plotz_width_dim,
39   width .initial:n = {0.75\linewidth},
40 }
41 \cs_new:Npn \__ztikz_pairs_domain_parse:w #1; #2\q_stop
42 {
43   \tl_set:Nn \l__pairs_x_domain_tl {#1}
44   \tl_if_empty:nTF {#2}
45     { \tl_set:Nn \l__pairs_y_domain_tl {*: *} }
46     { \tl_set:Nn \l__pairs_y_domain_tl {#2} }
```

```

47 }
48
49
50 % ==> plot precise
51 \bool_new:N \g__plot_precise_restore_bool
52 \bool_new:N \g__contour_precise_restore_bool
53 \bool_new:N \g__param_precise_restore_bool
54 \bool_new:N \g__polar_precise_restore_bool
55 \NewDocumentCommand\PlotPrecise{ smm }
56 {
57     \tl_if_in:nnF {plot, param, polar, contour}{#2}
58     {
59         \msg_new:nnn {ztikz}{ztikz-plot-type}
60         {Valid~plot~type~are:~'plot',~'param',~'polar'~and~'contour'}
61         \msg_error:nn {ztikz}{ztikz-plot-type}
62     }
63     \IfBooleanF{#1}{ \bool_gset_true:c {g__#2_precise_restore_bool} }
64     \tl_if_eq:nnTF {contour}{#2}{
65         \__ztikz_sed_script:nnn {contour_plot.gp}{2}{set~samples~#3}
66         \__ztikz_sed_script:nnn {contour_plot.gp}{3}{set~isosamples~#3,#3}
67     }{
68         \__ztikz_sed_script:nnn {#2_plot.gp}{3}{set~samples~#3}
69     }
70 }
71 % data plot, precise, style
72 \cs_new_protected:Npn \__ztikz_gnu_data_plot:nnn #1#2#3
73 {
74     \ztool_shell_escape:e {gnuplot~ \g__ztikz_scripts_path_tl/#1_plot.gp}
75     \int_gadd:Nn \g__gnu_data_index_int {1}
76     \tl_set:Ne \l__gnu_data_new_name_tl
77     {
78         gnu_data_\int_use:N \g__tikz_env_index_int
79         _\int_use:N \g__gnu_data_index_int.table
80     }
81     \tl_set:Ne \l__gnu_data_full_path_tl
82     { \g__ztikz_gnu_path_tl/\l__gnu_data_new_name_tl }
83     \ztool_shell_mv:ee
84     { \g__ztikz_gnu_path_tl/gnu_data.table }
85     { \l__gnu_data_full_path_tl }
86     \tl_if_empty:nTF {#3}
87     {
88         \draw[#2] plot[smooth] file {\l__gnu_data_full_path_tl};
89     }{
90         \group_begin:
91         \ztikz_keys_set:nn { point } { #3 }
92         \draw[#2] plot [
93             mark = \str_use:N \l__point_type_str,
94             mark~ size = \dim_use:N \l__point_radius_dim,

```

```

95         mark~ options = {
96             rotate = \fp_use:N \l__point_rotate_angle,
97             opacity = \tl_use:N \l__point_opacity_tl,
98             color = \tl_use:N \l__point_color_tl,
99             ball~ color = \tl_use:N \l__point_color_tl,
100         }
101     ] file {\l__gnu_data_full_path_tl};
102     \group_end:
103 }
104 \bool_if:cT {g__#1_precise_restore_bool}
105     { \PlotPrecise{#1}{100} }
106 }
107 \cs_generate_variant:Nn \__ztikz_gnu_data_plot:nnn {nee}
108
109 % ==> simple 2d function plot
110 \NewDocumentCommand\Plot{ 0{}m }
111 {
112     \group_begin:
113     \ztikz_keys_set:nn { 2dplot } { domain=-5:5, style={color=black}, marker= }
114     \ztikz_keys_set:nn { 2dplot } { #1 }
115     \__ztikz_sed_script:nne {plot_plot.gp}{8}{set~xr~[\l__ztikz_plot_domain_tl]}
116     \__ztikz_sed_script:nne {plot_plot.gp}{7}{f(x)~::~#2}
117     \__ztikz_gnu_data_plot:nee {plot}{\l__ztikz_plot_style_tl}{\l__ztikz_plot_marker_tl}
118     \group_end:
119 }
120
121 \NewDocumentCommand\ContourPlot{ 0{}m }
122 {
123     \group_begin:
124     \ztikz_keys_set:nn { 2dplot } { domain={-5:5}, style={color=black}, marker= }
125     \ztikz_keys_set:nn { 2dplot } { #1 }
126     \exp_last_unbraced:Nf \__ztikz_pairs_domain_parse:w \l__ztikz_plot_domain_tl\q_stop
127     \__ztikz_sed_script:nne {contour_plot.gp}{11}{set~xr~[\l__pairs_x_domain_tl]}
128     \__ztikz_sed_script:nne {contour_plot.gp}{12}{set~yr~[\l__pairs_y_domain_tl]}
129     \__ztikz_sed_script:nne {contour_plot.gp}{14}{f(x,~y)~::~#2}
130     \__ztikz_gnu_data_plot:nee {contour}{\l__ztikz_plot_style_tl}{\l__ztikz_plot_marker_tl}
131     \group_end:
132 }
133 \NewDocumentCommand\ParamPlot{ 0{domain=0:2*pi, style=black, marker=}m }
134 {
135     \group_begin:
136     \ztikz_keys_set:nn { 2dplot } { domain=0:2*pi, style=black, marker= }
137     \ztikz_keys_set:nn { 2dplot } { #1 }
138     \__ztikz_sed_script:nne {param_plot.gp}{8}{set~trange~[\l__ztikz_plot_domain_tl]}
139     \__ztikz_sed_script:nne {param_plot.gp}{9}{plot~#2}
140     \__ztikz_gnu_data_plot:nee {param}{\l__ztikz_plot_style_tl}{\l__ztikz_plot_marker_tl}
141     \group_end:
142 }

```

```

143 \NewDocumentCommand\PolarPlot{ 0{}m }
144 {
145     \group_begin:
146     \ztikz_keys_set:nn { 2dplot } { domain=0:2*pi, style=black, marker= }
147     \ztikz_keys_set:nn { 2dplot } { #1 }
148     \__ztikz_sed_script:nne {polar_plot.gp}{8}{set~trange~[\l__ztikz_plot_domain_tl]}
149     \__ztikz_sed_script:nne {polar_plot.gp}{9}{plot~#2}
150     \__ztikz_gnu_data_plot:nee {polar}{\l__ztikz_plot_style_tl}{\l__ztikz_plot_marker_tl}
151     \group_end:
152 }
153 \NewDocumentCommand\Plotz{ 0{}m }
154 {
155     \group_begin:
156     \ztikz_keys_set:nn { 3dplot } { #1 }
157     \bool_if:NTF \l__ztikz_plotz_pm_bool {\def\plotz@pm{with~pm3d}}{\def\plotz@pm{}}
158     \exp_last_unbraced:Nf \__ztikz_pairs_domain_parse:w \l__ztikz_plotz_domain_tl\q_stop
159     \__ztikz_sed_script:nne {3d_plot.gp}{18}{set~palette~\l__ztikz_plotz_palette_tl}
160     \__ztikz_sed_script:nne {3d_plot.gp}{23}{set~xr~[\l__pairs_x_domain_tl]}
161     \__ztikz_sed_script:nne {3d_plot.gp}{24}{set~yr~[\l__pairs_y_domain_tl]}
162     \__ztikz_sed_script:nne {3d_plot.gp}{25}{splot~#2~\plotz@pm}
163     \ztool_shell_escape:e {gnuplot~ ./ztikz_output/scripts/3d_plot.gp}
164     \tl_set:N \l__tmpa_tl {./ztikz_output/gnuplot_data/plot_3d_\int_use:N
\g__gnu_plotz_index_int.pdf}
165     \ztool_shell_mv:ne {./ztikz_output/gnuplot_data/plot_3d.pdf}{\l__tmpa_tl}
166     \includegraphics[width=\dim_use:N \l__ztikz_plotz_width_dim]{\l__tmpa_tl}
167     \int_gadd:Nn \g__gnu_plotz_index_int {1}
168     \group_end:
169 }
170
171
172 % ==> users' interface
173 \NewDocumentCommand{\currentTikzIndex}{}
174 {
175     \int_use:N \g__tikz_env_index_int
176 }
177 \def\gnudata#1
178 {
179     \tl_use:N \g__ztikz_gnu_path_tl/gnu_data_
180     \int_use:N \g__tikz_env_index_int _#1.table
181 }

```

8.2.3 cache

```
1 \ProvidesExplFile{ztikz.library.cache.tex}{2025/05/31}
2 {1.0.0}{cache~library~for~ztikz}
3
4
5
6 % ==> init cache
7 \clist_if_in:NnT \g__ztikz_library_loaded_clist {basic}
8 {
9   \ztool_shell_mkdir:n {ztikz_output/tikz_data/}
10   \usetikzlibrary{external}
11   \tikzexternalize[prefix=ztikz_output/tikz_data/]
12 }
13 \ztool_file_new:nn {\c_false_bool}{ztikz_output/ztikz.hash}
14
15
16 % ==> variables declaration
17 \ior_new:N \g__ztikz_file_ior
18 \tl_new:N \l__ztikz_current_hash_tl
19 \seq_new:N \g_ztikz_file_hash_seq
20 \seq_new:N \l__zcache_hash_label_seq
21 \seq_new:N \g__zcache_tmp_hash_seq
22 \tl_new:N \l__zcache_hash_label_tl
23 \tl_new:N \l__zcache_hash_hash_tl
24 \tl_new:N \g__zcache_latest_cache_label_tl
25 \bool_new:N \l__zcache_hash_label_miss_bool
26 \bool_new:N \g_ztikz_hash_nochg_run_bool
27 \bool_new:N \g_ztikz_hashchg_norun_bool
28 \bool_gset_false:N \g_ztikz_hashchg_norun_bool
29 \bool_gset_false:N \g_ztikz_hash_nochg_run_bool
30 \cs_generate_variant:Nn \ztikz_file_read_lines:n { e }
31
32
33 % ==> cache function
34 \prg_set_conditional:Npnn \ztikz_if_run_again:nnn #1#2#3 { p, T, F, TF }
35 {% #1:true/false; #2:True-->file, False-->str; #3:label
36   \tl_gset:Nn \g__zcache_latest_cache_label_tl { #3 }
37   \__zcache_hash_get:nn {#1} {#2}
38   \edef\zcache@flag
39   {
40     \tl_map_function:nN {
41       \g_ztikz_hashchg_norun_bool
42       \g_ztikz_hash_nochg_run_bool
43     } \int_eval:n
44   }
45   \exp_args:Ne \int_case:nnF
46   { \exp_not:N \int_from_bin:n {\zcache@flag} }
```

```

47 {
48 {0}{
49 \seq_if_in:NVTF \g_ztikz_file_hash_seq \l__ztikz_current_hash_tl
50 {
51 \ztikz_term_info:n {CURRENT~HASH~ALREADY~EXISTS}
52 \prg_return_false:
53 }{
54 \ztikz_term_info:n {CURRENT~HASH~IS~UNIQUE:RECORDING...}
55 \__zcache_hash_add:nn { #3 }{ \l__ztikz_current_hash_tl }
56 \prg_return_true:
57 }
58 }
59 {1}{
60 \ztikz_term_info:n {FORCE~TO~RUN~AGAIN~...}
61 \bool_gset_false:N \g_ztikz_hash_nochg_run_bool
62 \prg_return_true:
63 }
64 {2}{
65 \ztikz_term_info:n {FORCE~TO~SKIP~...}
66 \bool_gset_false:N \g_ztikz_hashchg_norun_bool
67 \__zcache_hash_extract_by_label:nnn
68 { ztikz_output/ztikz.hash }{ #3 }{-1}
69 \prg_return_false:
70 }
71 }{ \relax }
72 }
73 \cs_new_protected:Npn \__zcache_hash_get:nn #1#2
74 {% #1:true/false; #2:True-->file, False-->str;
75 \bool_if:nTF {#1}
76 { \file_get_md5five_hash:nN {#2} \l__ztikz_current_hash_tl }
77 { \tl_set:Nn \l__ztikz_current_hash_tl {#2} }
78 \tl_set_rescan:Nne \l__ztikz_current_hash_tl
79 { \cctab_select:N \c_initex_cctab }
80 { \l__ztikz_current_hash_tl }
81 \__zcache_hash_extract_all:nN
82 { ztikz_output/ztikz.hash }
83 \g_ztikz_file_hash_seq
84 \seq_gremove_duplicates:N \g_ztikz_file_hash_seq
85 \ztikz_term_info:e
86 {
87 \iow_newline:
88 CURRENT~FILE's~HASH:\l__ztikz_current_hash_tl
89 }
90 }
91 \cs_new_protected:Npn \__zcache_hash_extract_by_label:nnn #1#2#3
92 {% #1:file; #2:label; #3:index
93 \ztool_read_file_as_seq:nnN
94 { \c_false_bool }{ #1 }

```

```

95 \l_tmpa_seq
96 \bool_set_true:N \l__zcache_hash_label_miss_bool
97 \seq_map_inline:Nn \l_tmpa_seq
98 {
99   \zcache_hash_label_extract:nnN { label }{ ##1 }
100   \l__zcache_hash_label_tl
101   \zcache_hash_label_extract:nnN { hash }{ ##1 }
102   \l__zcache_hash_hash_tl
103   \tl_if_eq:NnT \l__zcache_hash_label_tl { #2 }
104   {
105     \bool_set_false:N \l__zcache_hash_label_miss_bool
106     \tl_set:Ne \l__ztikz_current_hash_tl
107     {
108       \clist_item:en {\l__zcache_hash_hash_tl}{#3}
109     }
110   }
111 }
112 \bool_if:NT \l__zcache_hash_label_miss_bool
113 {
114   \msg_set:nnn {ztikz}{hash-label-missing}
115   {
116     current~hash~label~'#2'~not~found~in~'ztikz.hash',~
117     do~NOT~change~the~hash~label~in~any~respect~after~cache.
118   }
119   \msg_error:nn {ztikz}{hash-label-missing}
120 }
121 }
122 \prg_generate_conditional_variant:Nnn \ztikz_if_run_again:nnn
123 { nen } { T, F, TF }
124
125 % extract hash or label
126 \cs_new_protected:Npn \__zcache_hash_add:nn #1#2
127 {% #1:label, #2:hash
128   \seq_clear:N \l_tmpa_seq
129   \seq_clear:N \l_tmpb_seq
130   \seq_clear:N \l__zcache_hash_label_seq
131   \ztool_read_file_as_seq:nnN
132   { \c_false_bool }
133   { ztikz_output/ztikz.hash }
134   \l_tmpa_seq
135   \seq_map_inline:Nn \l_tmpa_seq
136   {
137     \zcache_hash_label_extract:nnN { label }{ ##1 } \l_tmpa_tl
138     \seq_put_right:NV \l__zcache_hash_label_seq \l_tmpa_tl
139     \tl_if_eq:NnTF \l_tmpa_tl { #1 }
140     { \seq_put_right:Ne \l_tmpb_seq {##1, #2} }
141     { \seq_put_right:Ne \l_tmpb_seq {##1} }
142   }

```

```

143 \seq_if_in:NnF \l__zcache_hash_label_seq { #1 }
144 { \seq_put_right:Ne \l_tmpb_seq { #1:#2 } }
145 \ztool_write_seq_to_file:nNn { \c_true_bool }
146 \l_tmpb_seq { ztikz_output/ztikz.hash }
147 }
148 \cs_new_protected:Npn \__zcache_hash_extract_all:nN #1#2
149 {% #1:file; #2:seq
150 \seq_clear:N \l_tmpa_seq
151 \clist_clear:N \l_tmpa_clist
152 \ztool_read_file_as_seq:nnN
153 { \c_false_bool }{ #1 }
154 \l_tmpa_seq
155 \seq_map_inline:Nn \l_tmpa_seq
156 {
157 \zcache_hash_label_extract:nnN { hash }{ ##1 } \l_tmpa_tl
158 \seq_gset_from_clist:NN \g__zcache_tmp_hash_seq \l_tmpa_tl
159 \clist_put_right:NV \l_tmpa_clist \l_tmpa_tl
160 }
161 \seq_set_from_clist:NN \l_tmpb_seq \l_tmpa_clist
162 \seq_remove_duplicates:N \l_tmpb_seq
163 \seq_gset_eq:NN #2 \l_tmpb_seq
164 }
165 \cs_set:Npn \zcache_cache_hash_last:nnn #1#2#3
166 {% #1:file; #2:label; #3:index
167 \seq_clear:N \l_tmpa_seq
168 \clist_clear:N \l_tmpa_clist
169 \ztool_read_file_as_seq:nnN
170 { \c_false_bool }{ #1 }
171 \l_tmpa_seq
172 \seq_map_inline:Nn \l_tmpa_seq
173 {
174 \zcache_hash_label_extract:nnN { label }{ ##1 }
175 \l__zcache_hash_label_tl
176 \zcache_hash_label_extract:nnN { hash }{ ##1 }
177 \l__zcache_hash_hash_tl
178 \tl_if_eq:NnT \l__zcache_hash_label_tl { #2 }
179 {
180 \clist_item:en {\l__zcache_hash_hash_tl}{ #3 }
181 \seq_map_break:
182 }
183 }
184 }
185 \cs_generate_variant:Nn \zcache_cache_hash_last:nnn { nen, eee }
186 \ztikz_keys_define:nn { cache/hash }
187 {
188 label .tl_set:N = \l__cache_hash_label_user_tl,
189 label .initial:e = { \g__zcache_latest_cache_label_tl },
190 file .tl_set:N = \l__cache_hash_file_user_tl,

```

```

191 file .initial:e = { ztikz_output/ztikz.hash },
192 index .int_set:N = \l__cache_hash_index_user_int,
193 index .initial:n = { -1 },
194 }
195 \newcommand{\ztikzCachedHash}[1] []
196 {
197   \group_begin:
198     \ztikz_keys_set:nn { cache/hash } { #1 }
199     \zcache_cache_hash_last:eee
200     { \l__cache_hash_file_user_tl }
201     { \l__cache_hash_label_user_tl }
202     { \int_use:N \l__cache_hash_index_user_int }
203   \group_end:
204 }
205 \cs_set_protected:Npn \zcache_hash_label_extract:nnN #1#2#3
206 {
207   \tl_clear:N \l_tmpa_tl
208   \tl_set_rescan:Nnn \l_tmpa_tl
209   { \cctab_select:N \c_code_cctab }
210   { #2 }
211   \exp_args:NNe \tl_set:Ne #3
212   {
213     \exp_not:c {__hash_#1_extract:w}
214     \l_tmpa_tl \exp_not:N \q_stop
215   }
216 }
217 \cs_set:Npn \__hash_label_extract:w #1:#2\q_stop
218 { #1 }
219 \cs_set:Npn \__hash_hash_extract:w #1:#2\q_stop
220 { #2 }
221
222
223 % ==> clear cache hash
224 \cs_new_protected:Npn \ztikz_clear_hash:
225 {
226   \iow_open:Nn \g__ztikz_file_ior {ztikz_output/ztikz.hash}
227   \ior_close:N \g__ztikz_file_ior
228 }
229 \NewDocumentCommand{\ztikzHashClean}{s}{
230   \ztikz_clear_hash:
231   \ztikz_term_info:n {CLEAN~ALL~CACHED~HASH~SUCCESSFULLY...}
232 }
233 \NewDocumentCommand{\ztikzHashCurrent}{s+0{,}}{
234   \IfBooleanTF{#1}
235   { \tl_use:N \l__ztikz_current_hash_tl }
236   { \seq_use:Nn \g_ztikz_file_hash_seq {#2} }
237 }
238

```

239		239
240	% ==> override the cache mechanism	240
241	<code>\NewDocumentCommand{\ztikzForceToSkip}{}{</code>	241
242	<code>{</code>	242
243	<code>\bool_gset_false:N \g_ztikz_hash_nochg_run_bool</code>	243
244	<code>\bool_gset_true:N \g_ztikz_hashchg_norun_bool</code>	244
245	<code>}</code>	245
246	<code>\NewDocumentCommand{\ztikzForceToRun}{}{</code>	246
247	<code>{</code>	247
248	<code>\bool_gset_false:N \g_ztikz_hashchg_norun_bool</code>	248
249	<code>\bool_gset_true:N \g_ztikz_hash_nochg_run_bool</code>	249
250	<code>}</code>	250

8.2.4 python

```
1 \ProvidesExplFile{ztikz.library.python.tex}{2025/08/10}
2 {1.0.0}{python~library~for~ztikz}
3
4
5
6 % ==> writing scripts
7 \__ztikz_load_library:n {pyscript}
8 \ztool_shell_mkdir:n {ztikz_output/python_data/}
9 \tl_const:Nn \g__ztikz_python_path_tl {ztikz_output/python_data}
10 \ior_new:N \g__file_read_ior
11 \tl_new:N \g__file_content_tl
12
13
14 % ==> core functions
15 \cs_new_protected:Npn \zlatex_Readlines_cs:nn #1#2
16 {
17   \ior_open:Nn \g__file_read_ior {#2}
18   \str_case:nnF {#1}{
19     {raw}{
20       \ior_get:NN \g__file_read_ior \g__file_content_tl
21     }
22     {str}{
23       \ior_str_get:NN \g__file_read_ior \g__file_content_tl
24     }
25   }{}
26   \tl_use:N \g__file_content_tl
27 }
28 \cs_generate_variant:Nn \zlatex_Readlines_cs:nn {ee}
29
30
31 % ==> users' interface
32 % python-matplotlib
33 \NewDocumentEnvironment{pyfig}{ mm }
34 {% #1:label; #2:file name
35   \xsim_file_write_start:ne {\c_true_bool}{\g__ztikz_python_path_tl/t@mp.py}
36 }{
37   \xsim_file_write_stop:
38   \ztikz_if_run_again:nenTF {\c_true_bool}{\g__ztikz_python_path_tl/t@mp.py}{#1}
39   {
40     \__ztikz_addto_script:nn {\g__ztikz_python_path_tl/t@mp.py}
41     { plt.savefig('#2') }
42     \ztool_shell_escape:e {python~\g__ztikz_python_path_tl/t@mp.py}
43     \ztool_shell_mv:ee
44     { \g__ztikz_python_path_tl/t@mp.py }
45     { \g__ztikz_python_path_tl/pyfig_\l__ztikz_current_hash_tl.py }
46     \ztool_shell_mv:ee
```

```

47         { #2 }
48         { \g__ztikz_python_path_tl/\l__ztikz_current_hash_tl _#2 }
49         \ztikz_term_info:e
50         {
51             writing~source~to~file:'\g__ztikz_python_path_tl/
52             pyfig_\l__ztikz_current_hash_tl.py'
53         }
54     }{
55         \ztikz_term_info:e
56         {
57             skip~recompile~of~python,~use~the~
58             cache~picture:'\l__ztikz_current_hash_tl _#2'
59         }
60     }
61     \xdef\pyfigOutputFile{ \g__ztikz_python_path_tl/\l__ztikz_current_hash_tl _#2 }
62 }
63
64 % inline python command
65 \NewDocumentCommand\py{0{raw}m}
66 {
67     \__ztikz_sed_script:nne {python_script.py}{6}{Float_res~~~\tl_to_str:n {#2}}
68     \ztikz_term_info:e {using~python~float~module~calculating...}
69     \ztool_shell_escape:e {python~ \g__ztikz_scripts_path_tl/python_script.py}
70     \zlatex_Readlines_cs:ee {#1}{\g__ztikz_python_path_tl/PyFloat.out}
71     % ---> cause bug that can't write ToC to file
72     % \cs{iow_close:N} \cs{g__file_read_ior} leads to bug ??
73 }
74 % python-sympy
75 \NewDocumentCommand\sympy{ mm }
76 {
77     \__ztikz_sed_script:nne {sympy_script.py}{8}{F_res~~~\tl_to_str:n {#2}}
78     \ztikz_if_run_again:nenTF {\c_true_bool}{\g__ztikz_scripts_path_tl/sympy_script.py}{#1}
79     {
80         \ztool_shell_escape:e {python~ \g__ztikz_scripts_path_tl/sympy_script.py}
81         \ztool_shell_mv:ee
82         {\g__ztikz_python_path_tl/sympy.out}
83         {\g__ztikz_python_path_tl/sympy_\l__ztikz_current_hash_tl.out}
84         \ztikz_term_info:e {using~python~sympy~calculating~question~...}
85         \exp_args:Ne \input{\g__ztikz_python_path_tl/sympy_\l__ztikz_current_hash_tl.out}
86     }{
87         \exp_args:Ne \input{\g__ztikz_python_path_tl/sympy_\l__ztikz_current_hash_tl.out}
88         \ztikz_term_info:e {skip~recompile,~using~the~cache~sympy~result:~ ✓
89         \l__ztikz_current_hash_tl}
90     }
91 }
92 % python-code-env
93 \NewDocumentEnvironment{pycode}{ mm }
94 {
95     {% #1:label; #2:output file name (with ext)

```

94	\xsim_file_write_start:ne {\c_true_bool}{\g__ztikz_python_path_tl/t@mp.py}	94
95	}{	95
96	\xsim_file_write_stop:	96
97	\ztikz_if_run_again:nenTF {\c_true_bool}{\g__ztikz_python_path_tl/t@mp.py}{#1}	97
98	{	98
99	\ztool_shell_escape:e {python~\g__ztikz_python_path_tl/t@mp.py}	99
100	\ztool_shell_mv:ee	100
101	{ \g__ztikz_python_path_tl/t@mp.py }	101
102	{ \g__ztikz_python_path_tl/pycode_\l__ztikz_current_hash_tl.py }	102
103	\ztool_shell_mv:ee	103
104	{ #2 }	104
105	{ \g__ztikz_python_path_tl/\l__ztikz_current_hash_tl _#2 }	105
106	\ztikz_term_info:e	106
107	{	107
108	writing~source~to~file: '\g__ztikz_python_path_tl/	108
109	pycode_\l__ztikz_current_hash_tl.py'	109
110	}	110
111	}{	111
112	\ztikz_term_info:e	112
113	{	113
114	skip~recompile~of~python,~use~the~cache~result:	114
115	'\l__ztikz_current_hash_tl _#2'	115
116	}	116
117	}	117
118	\xdef\pycodeOutputFile{\g__ztikz_python_path_tl/\l__ztikz_current_hash_tl _#2}	118
119	}	119

8.2.5 wolfram

```
1 \ProvidesExplFile{ztikz.library.wolfram.tex}{2026/05/08}
2 {1.2.0}{wolfram~library~for~ztikz}
3
4
5
6 % ==> init variables
7 \ztool_shell_mkdir:n {ztikz_output/mma_data/}
8 \tl_const:Nn \g__ztikz_wolfram_path_tl {ztikz_output/mma_data}
9 \newcommand\wolframOutputPath{\g__ztikz_wolfram_path_tl}
10
11 \tl_new:N \l_part_table_data_tl
12 \tl_new:N \l_full_table_data_tl
13 \tl_new:N \l__wolfram_current_hash_tl
14 \tl_new:N \l__ztikz_wolfram_tmp_arg_tl
15 \tl_new:N \l__ztikz_wolfram_tmp_res_tl
16 \seq_new:N \g_ztikz_wolfram_res_seq
17 \ior_new:N \g__ztikz_wolfram_ior
18 \iow_new:N \g__ztikz_wolfram_iow
19
20
21 % ==> core function
22 \cs_new_protected:Npn \__ztikz_wolfram_execute:nnnn #1#2#3#4
23 {% #1:contents(empty->not add); #2:extension; #3:output object; #4:label
24   \__ztikz_wolfram_execute_prehook:nnn { #1 }{ #2 }{ #3 }
25   \__ztikz_wolfram_execute_cli:nn { #2 }{ #4 }
26   \__ztikz_wolfram_execute_posthook:n { #2 }
27 }
28 \cs_generate_variant:Nn \__ztikz_wolfram_execute:nnnn { e, o }
29
30 % wolfram execute aux functions
31 \cs_new:Npn \__ztikz_wolfram_execute_prehook:nnn #1#2#3
32 {
33   \edef\@wolfram@tmp@file{\g__ztikz_wolfram_path_tl/t@mp}
34
35   % add additional wolfram code - export process
36   \tl_if_empty:nF {#1}{
37     \__ztikz_addto_script:en { \@wolfram@tmp@file.wls }{ #1 }
38   }
39
40   % add additional wolfram code - script path
41   \tl_if_empty:nT { #3 }
42   {
43     \exp_args:Neee \ztool_insert_to_file:nnn
44       { \@wolfram@tmp@file.wls }{ 1 }
45       { scriptPath~=""\g__ztikz_wolfram_path_tl/" }
46   }
```

```

47 % get script hash
48 \file_get_md5five_hash:nN { \@wolfram@tmp@file.wls }\l__ztikz_current_hash_tl
49 \file_get_md5five_hash:nN { \@wolfram@tmp@file.wls }\l__wolfram_current_hash_tl
50 \tl_set_rescan:Nne \l__wolfram_current_hash_tl
51 { \cctab_select:N \c_initex_cctab }
52 { \l__wolfram_current_hash_tl }
53 \xdef\wolfram@tmp@file{\g__ztikz_wolfram_path_tl/\l__ztikz_current_hash_tl}
54 \ztool_shell_mv:ee {\@wolfram@tmp@file.wls}{\wolfram@tmp@file.wls}
55
56
57 % add additional wolfram code - general export
58 \tl_if_empty:nF { #3 }
59 {
60   \__ztikz_addto_script:ee { \wolfram@tmp@file.wls }
61   { Export["\wolfram@tmp@file.#2", #3] }
62   \xdef\wolframOutputFile {
63     \g__ztikz_wolfram_path_tl / \l__ztikz_current_hash_tl.#2
64   }
65 }
66 }
67
68 \cs_new:Npn \__ztikz_wolfram_execute_posthook:n #1
69 {
70   \ior_open:Ne \g__ztikz_wolfram_ior { \wolframOutputFile }
71   \ior_get:NN \g__ztikz_wolfram_ior \l__ztikz_wolfram_tmp_res_tl
72   \tl_set_rescan:Nno \l__ztikz_wolfram_tmp_res_tl
73   { \cctab_select:N \c_document_cctab }
74   { \l__ztikz_wolfram_tmp_res_tl }
75   \exp_args:NNe \seq_gset_split:NnV \g__ztikz_wolfram_res_seq
76   { \c_atsign_str } \l__ztikz_wolfram_tmp_res_tl
77   \ior_close:N \g__ztikz_wolfram_ior
78 }
79
80 \cs_new:Nn \__ztikz_wolfram_case_run:
81 {
82   \tl_if_eq:VnTF \g__ztikz_wolfram_engine_tl { wstp }
83   {
84     % NOTE: 'wstp' is much faster than the other engines
85     \edef\wolfram@script
86     {
87       wstp~ '<<< "\wolfram@tmp@file.wls"'
88     }
89   }
90   {
91     \edef\wolfram@script
92     {
93       \g__ztikz_wolfram_engine_tl\space
94       \bool_if:NT \g__ztikz_wolfram_cloud_bool {-cloud\space}

```

95	-script\space	95
96	\wolfram@tmp@file.wls	96
97	}	97
98	}	98
99	\ztikz_term_info:e	99
100	{	100
101	\g__ztikz_wolfram_engine_tl\space~is~running~on:	101
102	'\wolfram@tmp@file.wls'...\iow_newline:	102
103	}	103
104	\ztool_shell_escape:e { \wolfram@script }	104
105	}	105
106	\cs_new:Npn __ztikz_wolfram_case_skip:n #1	106
107	{% #1: ext	107
108	% remove the skip wolframscript file and result	108
109	\clist_if_in:NnT \g__ztikz_library_loaded_clist {cache}	109
110	{	110
111	\tl_if_eq:enT {\zcache@flag}{10}	111
112	{	112
113	\seq_if_in:NVF \g_ztikz_file_hash_seq \l__wolfram_current_hash_tl	113
114	{	114
115	\ztikz_term_info:e {Removing~result~of~hash:\l__wolfram_current_hash_tl}	115
116	\ztool_shell_rm:e {\wolfram@tmp@file.wls}	116
117	\ztool_shell_rm:e {\wolfram@tmp@file.#1}	117
118	}	118
119	}	119
120	}	120
121	\ztikz_term_info:e	121
122	{	122
123	Use~cache~result:'\g__ztikz_wolfram_path_tl/	123
124	\l__ztikz_current_hash_tl.#1'\iow_newline:	124
125	}	125
126	}	126
127	\cs_new:Npn __ztikz_wolfram_execute_cli:nn #1#2	127
128	{% #1: ext; #2: hash label	128
129	\ztikz_if_run_again:nenTF { \c_false_bool }{ \l__ztikz_current_hash_tl }{ #2 }	129
130	{	130
131	__ztikz_wolfram_case_run:	131
132	}{	132
133	__ztikz_wolfram_case_skip:n { #1 }	133
134	}	134
135	}	135
136		136
137		137
138	% ==> user interface	138
139	\msg_set:nnn {ztikz}{wolfram-arg_empty}	139
140	{ wolfram~library~error:~argument~of~command~'\c_backslash_str #1'~can~NOT~be~empty. }	140
141	\cs_new:Npn __ztikz_wolfram_arg_empty_assert:nn #1#2	141
142	{	142

```

143 \exp_args:No \tl_if_empty:nT { #2 }
144 {
145     \msg_error:nnn {ztikz}{wolfram-arg_empty}{#1}
146 }
147 }
148
149 % --> load result
150 \NewDocumentCommand\wolframResult{so}
151 {
152     \IfBooleanTF{#1}
153     {% must be integer expression, or it will raise bug.
154     \seq_item:Ne \g_ztikz_wolfram_res_seq
155     {\IfValueTF {#2}{\int_eval:n {#2}}{1}}
156 }{
157     \seq_use:Nn \g_ztikz_wolfram_res_seq
158     { \IfValueTF {#2}{#2}{,} }
159 }
160 }
161
162 % --> wolfram code
163 \NewDocumentEnvironment{wolframCode}{m}
164 {
165     \xsim_file_write_start:ne {\c_false_bool}{\g__ztikz_wolfram_path_tl/t@mp.wls}
166 }{
167     \xsim_file_write_stop:
168     \__ztikz_wolfram_execute:nnnn
169     {}{}{}{#1}
170 }
171
172 % --> wolfram graphicx
173 \NewDocumentEnvironment{wolframGraphics}{m0{}}
174 {
175     \gdef\zgraphics@spec{#2}
176     \xsim_file_write_start:ne {\c_false_bool}{\g__ztikz_wolfram_path_tl/t@mp.wls}
177 }{
178     \xsim_file_write_stop:
179     \__ztikz_wolfram_execute:nnnn {}{pdf}{FIGURE}{#1}
180     \tl_if_empty:eF {\zgraphics@spec}
181     {
182         \exp_after:wN \includegraphics \exp_after:wN
183         [\zgraphics@spec]{\wolfram@tmp@file.pdf}
184     }
185 }
186
187 % --> wolfram simple code
188 \NewDocumentCommand\wolfram{smm}
189 {
190     \__ztikz_wolfram_arg_empty_assert:nn {\wolfram}{#3}

```

```

191 \__ztikz_wolfram_execute:ennn
192 {
193 \IfBooleanTF{#1}
194 { TeXResult = ToString[#3]; }
195 { TeXResult = ToString[TeXForm[#3]]; }
196 }{txt}{TeXResult}{#2}
197 }
198
199 % --> wolfram tex code(expandable token replace)
200 \group_begin:
201 \char_set_catcode_escape:n { 36 }
202 \char_set_catcode_letter:n { 92 }
203 $cs_gset:Nn $__double_backslash:n
204 { $tl_if_eq:NNTF #1\_{\}\{#1} }
205 $gdef$wolframTex{
206 \char_set_catcode_letter:n { 92 }
207 $wolframTex@getarg
208 }
209 $gdef$wolframTex@getarg#1#2{
210 $__ztikz_wolfram_arg_empty_assert:nn {wolframTex}{#2}
211 $tl_set:Ne $l_tmpa_tl
212 {
213 \tl_map_function:nN {#2}
214 $__double_backslash:n
215 }
216 $__ztikz_wolfram_execute:onn
217 {TeXResult = TeXForm[ToExpression["$l_tmpa_tl", TeXForm]]}
218 {txt}{TeXResult}{#1}
219 \char_set_catcode_escape:n { 92 }
220 }
221 \char_set_catcode_escape:n { 92 }
222 \char_set_catcode_letter:n { 36 }
223 \group_end:
224
225 % --> wolfram table (extended the interface of 'latexalpha2')
226 \cs_set:Npn \__table_item_handle:n #1
227 {% the inner '\exp_not:N' prevent expansion from 'tabularray'.
228 \exp_not:n {
229 \exp_not:N \__wolfram_table_cell_cmd:n {#1}
230 },
231 }
232 \cs_set:Npn \__table_row_handle:n #1 % #1='{1, 2, 3}'
233 {
234 \clist_use:en
235 {
236 \clist_map_function:oN #1
237 \__table_item_handle:n
238 }{ & } \

```

239	}	239
240	\cs_generate_variant:Nn \clist_use:nn { en }	240
241	\cs_generate_variant:Nn \clist_map_function:nN { oN }	241
242	\cs_new:Npn __part_table_from_file:nN #1#2	242
243	{% #1:file; #2:data var	243
244	\ztool_gread_file_as_seq:neN {\c_true_bool}	244
245	{ #1 } \l_tmpa_seq	245
246	\tl_set:Ne #2	246
247	{	247
248	\seq_map_function:NN \l_tmpa_seq	248
249	__table_row_handle:n	249
250	}	250
251	}	251
252	\cs_set:Npn __full_table_from_file:nn #1#2	252
253	{% #1:file; #2:table header	253
254	__part_table_from_file:nN	254
255	{ #1 } \l_part_table_data_tl	255
256	\tl_set:Ne \l_full_table_data_tl	256
257	{	257
258	\tl_if_empty:eF {#2}{#2 \\}	258
259	\l_part_table_data_tl	259
260	}	260
261	\tl_set:Ne \l_part_table_data_tl	261
262	{ \l_part_table_data_tl }	262
263	}	263
264	\cs_generate_variant:Nn __full_table_from_file:nn { VV }	264
265	\cs_set:Npn __typeset_table:nnn #1#2#3	265
266	{% #1:table format; #2:table header; #3:table part data	266
267	\begin{tabular}{#1}	267
268	\hline	268
269	\bool_if:NT \l_wolfram_table_hdbt_rule_bool	269
270	{ #2\\ \hline }	270
271	#3	271
272	\hline	272
273	\end{tabular}	273
274	}	274
275	\cs_generate_variant:Nn __typeset_table:nnn { VVV }	275
276	\ztikz_keys_define:nn { wolfram / table }	276
277	{	277
278	format .tl_set:N = \l__ztikz_wolfram_table_format_tl,	278
279	format .initial:n = { *{12}{1} },	279
280	header .tl_set:N = \l__ztikz_wolfram_table_header_tl,	280
281	header .initial:n = { },	281
282	hdbt-rule .bool_set:N = \l_wolfram_table_hdbt_rule_bool,	282
283	hdbt-rule .initial:n = { false },	283
284	hdbt-rule .default:n = { true },	284
285	cell-cmd .cs_gset:Np = __wolfram_table_cell_cmd:n #1,	285
286	cell-cmd .initial:n = { #1 },	286

```

287 }
288 \NewDocumentCommand{\wolframTable}{sm0{m}}
289 {% #1:if typeset; #2:key-value; #3:code
290   \__ztikz_wolfram_arg_empty_assert:nn { wolframTable }{ #4 }
291   \group_begin:
292     \ztikz_keys_set:nn { wolfram/table }{ #3 }
293     \__ztikz_wolfram_execute:ennn
294       { TeXResult = #4; }{ txt }
295       { TeXResult }{ #2 }
296     \__full_table_from_file:VV \wolframOutputFile
297     \l__ztikz_wolfram_table_header_tl
298     \IfBooleanT{#1}
299     {
300       \__typeset_table:VVV
301       \l__ztikz_wolfram_table_format_tl
302       \l__ztikz_wolfram_table_header_tl
303       \l_part_table_data_tl
304     }
305     \exp_args:NNo \gdef\wolframTablePData{ \l_part_table_data_tl }
306     \exp_args:NNo \gdef\wolframTableFData{ \l_full_table_data_tl }
307   \group_end:
308 }
309
310 % --> equation solve
311 \ztikz_keys_define:nn { wolfram/solve }
312 {
313   var .tl_set:N = \l__ztikz_wolfram_var_tl,
314   var .initial:n = {},
315   domain .tl_set:N = \l__ztikz_wolfram_domain_tl,
316   domain .initial:n = {},
317 }
318 \NewDocumentCommand\wolframSolve{smom}
319 {
320   \__ztikz_wolfram_arg_empty_assert:nn { wolframSolve }{ #4 }
321   \group_begin:
322   \IfValueT {#3} { \ztikz_keys_set:nn { wolfram/solve } {#3} }
323   \tl_if_empty:VF \l__ztikz_wolfram_domain_tl
324     { \tl_set:N \l__ztikz_wolfram_tmp_arg_tl {,\l__ztikz_wolfram_domain_tl} }
325   \__ztikz_wolfram_execute:ennn
326   {
327     \IfBooleanTF {#1}{
328       TeXResult = Row[Solve[#4]//Flatten, "@"]
329       /.{Rule -> Equal}//TeXForm//ToString;
330     }{
331       TeXResult = Row[
332         Solve[#4, {\l__ztikz_wolfram_var_tl} \l__ztikz_wolfram_tmp_arg_tl]//Flatten,
333         "@"
334       ]/.{Rule -> Equal}//TeXForm//ToString;

```

335	}	335
336	}{txt}{TeXResult}{#2}	336
337	\group_end:	337
338	}	338
339		339
340	% --> differential equation solve	340
341	\ztikz_keys_define:nn { wolfram/dsolve }	341
342	{	342
343	depend .tl_set:N = \l__ztikz_wolfram_de_var_tl,	343
344	depend .initial:n = { y[x] },	344
345	independ .tl_set:N = \l__ztikz_wolfram_in_var_tl,	345
346	independ .initial:n = { x },	346
347	}	347
348	\NewDocumentCommand\wolframDSolve{smom}	348
349	{	349
350	__ztikz_wolfram_arg_empty_assert:nn { wolframDSolve }{ #4 }	350
351	\group_begin:	351
352	\IfValueT {#3} { \ztikz_keys_set:nn { wolfram/dsolve } {#3} }	352
353	\tl_if_empty:VF \l__ztikz_wolfram_in_var_tl	353
354	{ \tl_set:Nx \l__ztikz_wolfram_in_var_tl {,\l__ztikz_wolfram_in_var_tl} }	354
355	__ztikz_wolfram_execute:ennn	355
356	{	356
357	\IfBooleanTF {#1}{	357
358	TeXResult = Row[DSolve[#4]//Flatten, ","]	358
359	/.{Rule -> Equal}//TeXForm//ToString;	359
360	}{	360
361	TeXResult = Row[	361
362	DSolve[{#4}, {\l__ztikz_wolfram_de_var_tl}\l__ztikz_wolfram_in_var_tl]//Flatten,	362
363	"@"	363
364	]/.{Rule -> Equal}//TeXForm//ToString;	364
365	}	365
366	}{txt}{TeXResult}{#2}	366
367	\group_end:	367
368	}	368

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